13 Correlation and Linear Regression

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TRAVELAIR.COM samples domestic airline flights to explore the relationship between airfare and distance. The service would like to know if there is a correlation between airfare and flight distance. If there is a correlation, what percentage of the variation in airfare is accounted for by distance? How much does each additional mile add to the fare? (See Exercise 61 and LO13-2, LO13-3, and LO13-5.)

LEARNING OBJECTIVES

When you have completed this chapter, you will be able to:

LO13-1 Explain the purpose of correlation analysis.

LO13-2 Calculate a correlation coefficient to test and interpret the relationship between two variables.

LO13-3 Apply regression analysis to estimate the linear relationship between two variables.

LO13-4 Evaluate the significance of the slope of the regression equation.

LO13-5 Evaluate a regression equation's ability to predict using the standard estimate of the error and the coefficient of determination.

LO13-6 Calculate and interpret confidence and prediction intervals.

LO13-7 Use a log function to transform a nonlinear relationship.

INTRODUCTION

Chapters 2 through 4 presented *descriptive statistics.* We organized raw data into a frequency distribution and computed several measures of location and measures of dispersion to describe the major characteristics of the distribution. In Chapters 5 through 7, we described probability, and from probability statements, we created probability distributions. In Chapters 8 through 12, we studied *statistical inference,* where we collected a sample to estimate a population parameter such as the population mean or population proportion. In addition, we used the sample data to test a hypothesis about a population mean or a population proportion, the difference between two population means, or the equality of several population means. Each of these tests involved just *one* interval- or ratio-level variable, such as the profit made on a car sale, the income of bank presidents, or the number of patients admitted each month to a particular hospital.

In this chapter, we shift the emphasis to the study of relationships between two interval- or ratio-level variables. In all business fields, identifying and studying relationships between variables can provide information on ways to increase profits, methods to decrease costs, or variables to predict demand. In marketing products, many firms use price reductions through coupons and discount pricing to increase sales. In this example, we are interested in the relationship between two variables: price reductions and sales. To collect the data, a company can test-market a variety of price reduction methods and observe sales. We hope to confirm a relationship that decreasing price leads to increased sales. In economics, you will find many relationships between two variables that are the basis of economics, such as price and demand.

STATISTICS IN ACTION

The space shuttle *Challenger* exploded on January 28, 1986. An investigation of the cause examined four contractors: Rockwell International for the shuttle and engines, Lockheed Martin for ground support, Martin Marietta for the external fuel tanks, and Morton Thiokol for the solid fuel booster rockets. After several months, the investigation blamed the explosion on defective O-rings produced by Morton Thiokol. A study of the contractor's stock prices showed an interesting happenstance. On the day of the *Challenger* explosion, Morton Thiokol stock was down 11.86% and the stock of the other three lost only 2 to 3%. Can we conclude that financial markets predicted the outcome of the investigation?

As another familiar example, recall in Chapter 4 we used the Applewood Auto Group data to show the relationship between two variables with a scatter diagram. We plotted the profit for each vehicle sold on the vertical axis and the age of the buyer on the horizontal axis. See page 116. In that graph, we observed that as the age of the buyer increased, the profit for each vehicle also increased.

Other examples of relationships between two variables are:

- Does the amount Healthtex spends per month on training its sales force affect its monthly sales?
- Is the number of square feet in a home related to the cost to heat the home in January?
- In a study of fuel efficiency, is there a relationship between miles per gallon and the weight of a car?
- Does the number of hours that students study for an exam influence the exam score?

In this chapter, we carry this idea further. That is, we develop numerical measures to express the relationship between two variables. Is the relationship strong or weak? Is it direct or inverse? In addition, we develop an equation to express the relationship between variables. This will allow us to estimate one variable on the basis of another.

To begin our study of relationships between two variables, we examine the meaning and purpose of **correlation analysis.** We continue by developing an equation that will allow us to estimate the value of one variable based on the value of another. This is called **regression analysis.** We will also evaluate the ability of the equation to accurately make estimations.

LO13-1
Explain the purpose of correlation analysis.

WHAT IS CORRELATION ANALYSIS?

When we study the relationship between two interval- or ratio-scale variables, we often start with a scatter diagram. This procedure provides a visual representation of the relationship between the variables. The next step is usually to calculate the correlation coefficient. It provides a quantitative measure of the strength of the relationship between

TABLE 13–1 Number of Sales Calls and Copiers Sold for 15 Salespeople

Sales Representative	Sales Calls	Copiers Sold
Brian Virost	96	41
Carlos Ramirez	40	41
Carol Saia	104	51
Greg Fish	128	60
Jeff Hall	164	61
Mark Reynolds	76	29
Meryl Rumsey	72	39
Mike Kiel	80	50
Ray Snarsky	36	28
Rich Niles	84	43
Ron Broderick	180	70
Sal Spina	132	56
Soni Jones	120	45
Susan Welch	44	31
Tom Keller	84	30

two variables. As an example, the sales manager of North American Copier Sales, which has a large sales force throughout the United States and Canada, wants to determine whether there is a relationship between the number of sales calls made in a month and the number of copiers sold that month. The manager selects a random sample of 15 representatives and determines, for each representative, the number of sales calls made and the number of copiers sold. This information is reported in Table 13–1.

By reviewing the data, we observe that there does seem to be some relationship between the number of sales calls and the number of units sold. That is, the salespeople who made the most sales calls sold the most units. However, the relationship is not "perfect" or exact. For example, Soni Jones made fewer sales calls than Jeff Hall, but she sold more units.

In addition to the graphical techniques in Chapter 4, we will develop numerical measures to precisely describe the relationship between the two variables, sales calls and copiers sold. This group of statistical techniques is called **correlation analysis.**

CORRELATION ANALYSIS A group of techniques to measure the relationship between two variables.

The basic idea of correlation analysis is to report the relationship between two variables. The usual first step is to plot the data in a **scatter diagram.** An example will show how a scatter diagram is used.

EXAMPLE

North American Copier Sales sells copiers to businesses of all sizes throughout the United States and Canada. Ms. Marcy Bancer was recently promoted to the position of national sales manager. At the upcoming sales meeting, the sales representatives from all over the country will be in attendance. She would like to impress upon them the importance of making that extra sales call each day. She decides to gather some information on the relationship between the number of sales calls and the number of copiers sold. She selects a random sample of 15 sales representatives and determines the number of sales calls they made last month and the number of copiers they sold. The sample information is reported in Table 13–1. What observations can you make about the relationship between the number of sales calls and the number of copiers sold? Develop a scatter diagram to display the information.

SOLUTION

Based on the information in Table 13–1, Ms. Bancer suspects there is a relationship between the number of sales calls made in a month and the number of copiers sold. Ron Broderick sold the most copiers last month and made 180 sales calls. On the other hand, Ray Snarsky, Carlos Ramirez, and Susan Welch made the fewest calls: 36, 40, and 44. They also had the lowest number of copiers sold among the sampled representatives.

The implication is that the number of copiers sold is related to the number of sales calls made. As the number of sales calls increases, it appears the number of copiers sold also increases. We refer to number of sales calls as the **independent variable** and number of copiers sold as the **dependent variable.**

The independent variable provides the basis for estimating or predicting the dependent variable. For example, we would like to predict the expected number of copiers sold if a salesperson makes 100 sales calls. In the randomly selected sample data, the independent variable—sales calls—is a random number.

The dependent variable is the variable that is being predicted or estimated. It can also be described as the result or outcome for a particular value of the independent variable. The dependent variable is random. That is, for a given value of the independent variable, there are many possible outcomes for the dependent variable.

It is common practice to scale the dependent variable (copiers sold) on the vertical or *Y*-axis and the independent variable (number of sales calls) on the horizontal or *X*-axis. To develop the scatter diagram of the North American Copier Sales information, we begin with the first sales representative, Brian Virost. Brian made 96 sales calls last month and sold 41 copiers, so $x = 96$ and $y = 41$. To plot this point, move along the horizontal axis to $x = 96$, then go vertically to $y = 41$ and place a dot at the intersection. This process is continued until all the paired data are plotted, as shown in Chart 13–1.

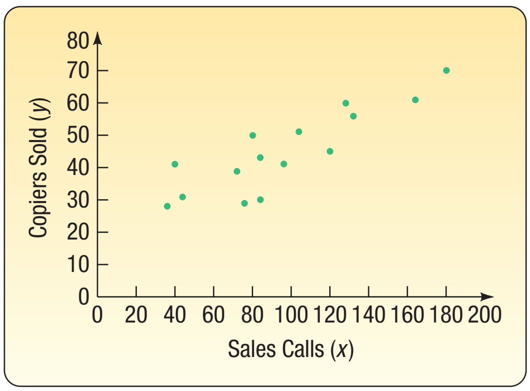


CHART 13–1 Scatter Diagram Showing Sales Calls and Copiers Sold

The scatter diagram shows graphically that the sales representatives who make more calls tend to sell more copiers. It is reasonable for Ms. Bancer, the national sales manager, to tell her salespeople that the more sales calls they make, the more copiers they can expect to sell. Note that, while there appears to be a positive relationship between the two variables, all the points do not fall on a straight line. In the following section, you will measure the strength and direction of this relationship between two variables by determining the correlation coefficient.

LO13-2
Calculate a correlation coefficient to test and interpret the relationship between two variables.

THE CORRELATION COEFFICIENT

Originated by Karl Pearson about 1900, the **correlation coefficient** describes the strength of the relationship between two sets of interval-scaled or ratio-scaled variables. Designated *r,* it is often referred to as *Pearson's r* and as the *Pearson product-moment correlation coefficient*. It can assume any value from −1.00 to +1.00 inclusive. A correlation coefficient of −1.00 or +1.00 indicates *perfect correlation*. For example, a correlation coefficient for the preceding example computed to be +1.00 would indicate that the number of sales calls and the number of copiers sold are perfectly related in a positive linear sense. A computed value of −1.00 would reveal that sales calls and the number of copiers sold are perfectly related in an inverse linear sense. How the scatter diagram would appear if the relationship between the two variables were linear and perfect is shown in Chart 13–2.

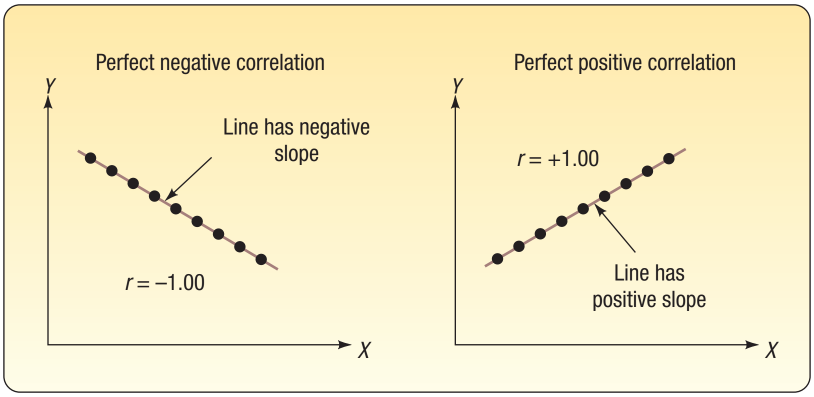


CHART 13–2 Scatter Diagrams Showing Perfect Negative Correlation and Perfect Positive Correlation

If there is absolutely no relationship between the two sets of variables, Pearson's *r* is zero. A correlation coefficient *r* close to 0 (say, .08) shows that the linear relationship is quite weak. The same conclusion is drawn if $r = -.08$. Coefficients of −.91 and +.91 have equal strength; both indicate very strong correlation between the two variables. Thus, *the strength of the correlation does not depend on the direction (either − or +).*

Scatter diagrams for $r = 0$, a weak *r* (say, −.23), and a strong *r* (say, +.87) are shown in Chart 13–3. Note that, if the correlation is weak, there is considerable scatter about a line drawn through the center of the data. For the scatter diagram representing a strong relationship, there is very little scatter about the line. This indicates, in the example shown on the chart, that hours studied is a good predictor of exam score.

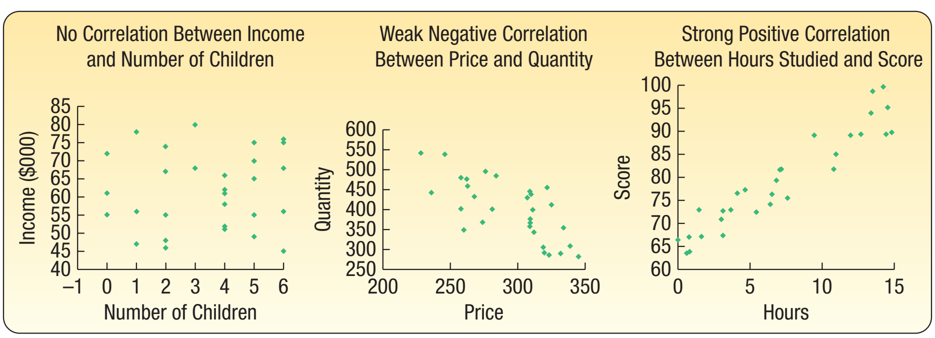
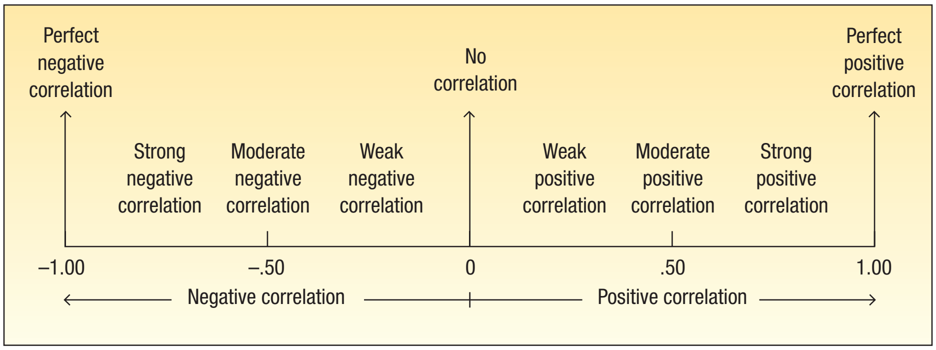


CHART 13–3 Scatter Diagrams Depicting Zero, Weak, and Strong Correlation

The following drawing summarizes the strength and direction of the correlation coefficient.



CORRELATION COEFFICIENT A measure of the strength of the linear relationship between two variables.

The characteristics of the correlation coefficient are summarized below.

CHARACTERISTICS OF THE CORRELATION COEFFICIENT

1. The sample correlation coefficient is identified by the lowercase letter r.
2. It shows the direction and strength of the linear relationship between two interval- or ratio-scale variables.
3. It ranges from -1 up to and including $+1$.
4. A value near 0 indicates there is little linear relationship between the variables.
5. A value near 1 indicates a direct or positive linear relationship between the variables.
6. A value near -1 indicates an inverse or negative linear relationship between the variables.

How is the value of the correlation coefficient determined? We will use the North American Copier Sales in Table 13–1 as an example. It is replicated in Table 13–2 for your convenience.

TABLE 13–2 Number of Sales Calls and Copiers Sold for 15 Salespeople

Sales Representative	Sales Calls	Copiers Sold
Brian Virost	96	41
Carlos Ramirez	40	41
Carol Saia	104	51
Greg Fish	128	60
Jeff Hall	164	61
Mark Reynolds	76	29
Meryl Rumsey	72	39
Mike Kiel	80	50
Ray Snarsky	36	28
Rich Niles	84	43
Ron Broderick	180	70
Sal Spina	132	56
Soni Jones	120	45
Susan Welch	44	31
Tom Keller	84	30
Total	1440	675

We begin with a scatter diagram, similar to Chart 13–2. Draw a vertical line through the data values at the mean of the *x*-values and a horizontal line at the mean of the *y*-values. In Chart 13–4, we've added a vertical line at 96 calls ($\bar{x} = \Sigma x/n = 1440/15 = 96$) and a horizontal line at 45 copiers ($\bar{y} = \Sigma y/n = 675/15 = 45$). These lines pass through the "center" of the data and divide the scatter diagram into four quadrants. Think of moving the origin from (0, 0) to (96, 45).

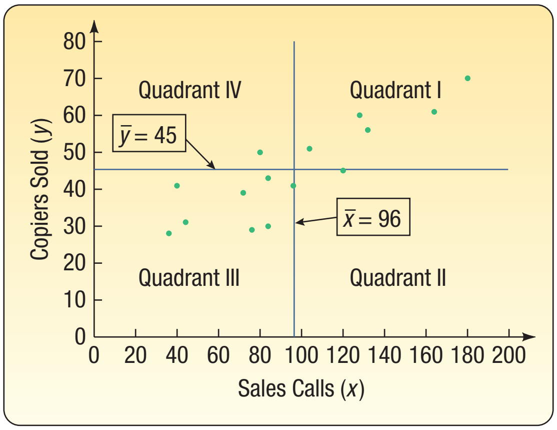


CHART 13–4 Computation of the Correlation Coefficient

Two variables are positively related when the number of copiers sold is above the mean and the number of sales calls is also above the mean. These points appear in the upper-right quadrant (labeled Quadrant I) of Chart 13–4. Similarly, when the number of copiers sold is less than the mean, so is the number of sales calls. These points fall in the lower-left quadrant of Chart 13–4 (labeled Quadrant III). For example, the third person on the list in Table 13–2, Carol Saia, made 104 sales calls and sold 51 copiers. These values are above their respective means, so this point is located in Quadrant I, which is in the upper-right quadrant. She made 8 more calls than the mean number of sales calls and sold 6 more than the mean number sold. Tom Keller, the last name on the list in Table 13–2, made 84 sales calls and sold 30 copiers. Both of these values are less than their respective means, hence this point is in the lower-left quadrant. Tom made 12 fewer sales calls and sold 15 fewer copiers than the respective means. The deviations from the mean number of sales calls and the mean number of copiers sold are summarized in Table 13–3 for the

TABLE 13–3 Deviations from the Mean and Their Products

Sales Representative	Sales Calls (x)	Copiers Sold (y)	$x - \bar{x}$	$x - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
Brian Virost	96	41	0	−4	0
Carlos Ramirez	40	41	−56	−4	224
Carol Saia	104	51	8	6	48
Greg Fish	128	60	32	15	480
Jeff Hall	164	61	68	16	1,088
Mark Reynolds	76	29	−20	−16	320
Meryl Rumsey	72	39	−24	−6	144
Mike Kiel	80	50	−16	5	−80
Ray Snarsky	36	28	−60	−17	1,020
Rich Niles	84	43	−12	−2	24
Ron Broderick	180	70	84	25	2,100
Sal Spina	132	56	36	11	396
Soni Jones	120	45	24	0	0
Susan Welch	44	31	−52	−14	728
Tom Keller	84	30	−12	−15	180
Totals	1440	675	0	0	6,672

15 sales representatives. The sum of the products of the deviations from the respective means is 6672. That is, the term $\Sigma(x - \bar{x})(y - \bar{y}) = 6672$.

In both the upper-right and the lower-left quadrants, the product of $(x - \bar{x})(y - \bar{y})$ is positive because both of the factors have the same sign. In our example, this happens for all sales representatives except Mike Kiel. Mike made 80 sales calls (which is less than the mean) but sold 50 machines (which is more than the mean). We can therefore expect the correlation coefficient to have a positive value.

If the two variables are inversely related, one variable will be above the mean and the other below the mean. Most of the points in this case occur in the upper-left and lower-right quadrants, that is, Quadrants II and IV. Now $(x - \bar{x})$ and $(y - \bar{y})$ will have opposite signs, so their product is negative. The resulting correlation coefficient is negative.

What happens if there is no linear relationship between the two variables? The points in the scatter diagram will appear in all four quadrants. The negative products of $(x - \bar{x})(y - \bar{y})$ offset the positive products, so the sum is near zero. This leads to a correlation coefficient near zero. So, the term $\Sigma (x - \bar{x})(y - \bar{y})$ drives the strength as well as the sign of the relationship between the two variables.

The correlation coefficient is also unaffected by the units of the two variables. For example, if we had used hundreds of copiers sold instead of the number sold, the correlation coefficient would be the same. The correlation coefficient is independent of the scale used if we divide the term $\Sigma (x - \bar{x})(y - \bar{y})$ by the sample standard deviations. It is also made independent of the sample size and bounded by the values +1.00 and −1.00 if we divide by $(n - 1)$.

This reasoning leads to the following formula:

CORRELATION COEFFICIENT

$$r = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{(n - 1)s_x s_y} \qquad \textbf{[13–1]}$$

To compute the correlation coefficient, we use the standard deviations of the sample of 15 sales calls and 15 copiers sold. We could use formula (3–9) to calculate the sample standard deviations or we could use a statistical software package. For the specific Excel and Minitab commands, see the **Software Commands** in Appendix C. The following is the Excel output. The standard deviation of the number of sales calls is 42.76 and of the number of copiers sold 12.89.

	A	B	C	D	E	F	G	H
1		Sales Representative	Sales Calls (x)	Copiers Sold (y)			Sales Calls (x)	Copiers Sold (y)
2		Brian Virost	96	41		Mean	96.00	45.00
3		Carlos Ramirez	40	41		Standard Error	11.04	3.33
4		Carol Saia	104	51		Median	84.00	43.00
5		Greg Fish	128	60		Mode	84.00	41.00
6		Jeff Hall	164	61		Standard Deviation	42.76	12.89
7		Mark Reynolds	76	29		Sample Variance	1828.57	166.14
8		Meryl Rumsey	72	39		Kurtosis	-0.32	-0.73
9		Mike Kiel	80	50		Skewness	0.46	0.36
10		Ray Snarsky	36	28		Range	144.00	42.00
11		Rich Niles	84	43		Minimum	36.00	28.00
12		Ron Broderick	180	70		Maximum	180.00	70.00
13		Sal Spina	132	56		Sum	1440.00	675.00
14		Soni Jones	120	45		Count	15.00	15.00
15		Susan Welch	44	31				
16		Tom Keller	84	30				
17		Total	1440	675				

We now insert these values into formula (13–1) to determine the correlation coefficient:

$$r = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{(n - 1)s_x s_y} = \frac{6672}{(15 - 1)(42.76)(12.89)} = 0.865$$

How do we interpret a correlation of 0.865? First, it is positive, so we conclude there is a direct relationship between the number of sales calls and the number of copiers sold. This confirms our reasoning based on the scatter diagram, Chart 13–4. The value of 0.865 is fairly close to 1.00, so we conclude that the association is strong.

We must be careful with the interpretation. The correlation of 0.865 indicates a strong positive linear association between the variables. Ms. Bancer would be correct to encourage the sales personnel to make that extra sales call because the number of sales calls made is related to the number of copiers sold. However, does this mean that more sales calls *cause* more sales? No, we have not demonstrated cause and effect here, only that the two variables—sales calls and copiers sold—are statistically related.

If there is a strong relationship (say, .97) between two variables, we are tempted to assume that an increase or decrease in one variable *causes* a change in the other variable. For example, historically, the consumption of Georgia peanuts and the consumption of aspirin have a strong correlation. However, this does not indicate that an increase in the consumption of peanuts *caused* the consumption of aspirin to increase. Likewise, the incomes of professors and the number of inmates in mental institutions have increased proportionately. Further, as the population of donkeys has decreased, there has been an increase in the number of doctoral degrees granted. Relationships such as these are called **spurious correlations.** What we can conclude when we find two variables with a strong correlation is that there is a relationship or association between the two variables, not that a change in one causes a change in the other.

EXAMPLE

The Applewood Auto Group's marketing department believes younger buyers purchase vehicles on which lower profits are earned and the older buyers purchase vehicles on which higher profits are earned. They would like to use this information as part of an upcoming advertising campaign to try to attract older buyers, for whom the profits tend to be higher. Develop a scatter diagram depicting the relationship between vehicle profits and age of the buyer. Use statistical software to determine the correlation coefficient. Would this be a useful advertising feature?

SOLUTION

Using the Applewood Auto Group example, the first step is to graph the data using a scatter plot. It is shown in Chart 13–5.

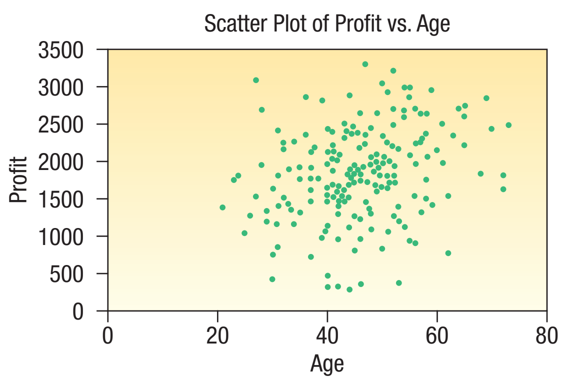


CHART 13–5 Scatter Diagram of Profit versus Age for the Applewood Auto Group Data

The scatter diagram suggests that a positive relationship does exist between age and profit; however, that relationship does not appear strong.

The next step is to calculate the correlation coefficient to evaluate the relative strength of the relationship. Statistical software provides an easy way to calculate the value of the correlation coefficient. The Excel output follows.

	A	B	C
1		***Age***	***Profit***
2	Age	1	
3	Profit	0.262	1

For these data, $r = 0.262$. To evaluate the relationship between a buyer's age and the profit on a car sale:

1. The relationship is positive or direct. Why? Because the sign of the correlation coefficient is positive. This confirms that as the age of the buyer increases, the profit on a car sale also increases.
2. The correlation coefficient is: $r = 0.262$. It is much closer to zero than one. Therefore, the relationship between the two variables is weak. We would observe that the relationship between the age of a buyer and the profit of their purchase is not very strong.

For Applewood Auto Group, the data does not support a business decision to create an advertising campaign to attract older buyers.

SELF-REVIEW 13–1

Haverty's Furniture is a family business that has been selling to retail customers in the Chicago area for many years. The company advertises extensively on radio, TV, and the Internet, emphasizing low prices and easy credit terms. The owner would like to review the relationship between sales and the amount spent on advertising. Below is information on sales and advertising expense for the last four months.

Month	Advertising Expense ($ million)	Sales Revenue ($ million)
July	2	7
August	1	3
September	3	8
October	4	10

(a) The owner wants to forecast sales on the basis of advertising expense. Which variable is the dependent variable? Which variable is the independent variable?
(b) Draw a scatter diagram.
(c) Determine the correlation coefficient.
(d) Interpret the strength of the correlation coefficient.

EXERCISES

1. **FILE** The following sample of observations were randomly selected.

x	4	5	3	6	10
y	4	6	5	7	7

Determine the correlation coefficient and interpret the relationship between x and y.

2. **FILE** The following sample of observations were randomly selected.

x	5	3	6	3	4	4	6	8
y	13	15	7	12	13	11	9	5

Determine the correlation coefficient and interpret the relationship between x and y.

3. **FILE** Bi-lo Appliance Super-Store has outlets in several large metropolitan areas in New England. The general sales manager aired a commercial for a digital camera on selected local TV stations prior to a sale starting on Saturday and ending Sunday. She obtained the information for Saturday–Sunday digital camera sales at the various outlets and paired it with the number of times the advertisement was shown on the local TV stations. The purpose is to find whether there is any relationship between the number of times the advertisement was aired and digital camera sales. The pairings are:

Location of TV Station	Number of Airings	Saturday–Sunday Sales ($ thousands)
Providence	4	15
Springfield	2	8
New Haven	5	21
Boston	6	24
Hartford	3	17

   a. What is the dependent variable?
   b. Draw a scatter diagram.
   c. Determine the correlation coefficient.
   d. Interpret these statistical measures.

4. **FILE** The production department of Celltronics International wants to explore the relationship between the number of employees who assemble a subassembly and the number produced. As an experiment, two employees were assigned to assemble the subassemblies. They produced 15 during a one-hour period. Then four employees assembled them. They produced 25 during a one-hour period. The complete set of paired observations follows.

Number of Assemblers	One-Hour Production (units)
2	15
4	25
1	10
5	40
3	30

The dependent variable is production; that is, it is assumed that different levels of production result from a different number of employees.

   a. Draw a scatter diagram.
   b. Based on the scatter diagram, does there appear to be any relationship between the number of assemblers and production? Explain.
   c. Compute the correlation coefficient.

5. **FILE** The city council of Pine Bluffs is considering increasing the number of police in an effort to reduce crime. Before making a final decision, the council asked the chief of police to survey other cities of similar size to determine the relationship between the number of police and the number of crimes reported. The chief gathered the following sample information.

City	Police	Number of Crimes	City	Police	Number of Crimes
Oxford	15	17	Holgate	17	7
Starksville	17	13	Carey	12	21
Danville	25	5	Whistler	11	19
Athens	27	7	Woodville	22	6

a. Which variable is the dependent variable and which is the independent variable? Hint: Which of the following makes better sense: Cities with more police have fewer crimes, or cities with fewer crimes have more police? Explain your choice.
b. Draw a scatter diagram.
c. Determine the correlation coefficient.
d. Interpret the correlation coefficient. Does it surprise you that the correlation coefficient is negative?

6. **FILE** The owner of Maumee Ford-Volvo wants to study the relationship between the age of a car and its selling price. Listed below is a random sample of 12 used cars sold at the dealership during the last year.

Car	Age (years)	Selling Price ($000)	Car	Age (years)	Selling Price ($000)
1	9	8.1	7	8	7.6
2	7	6.0	8	11	8.0
3	11	3.6	9	10	8.0
4	12	4.0	10	12	6.0
5	8	5.0	11	6	8.6
6	7	10.0	12	6	8.0

a. Draw a scatter diagram.
b. Determine the correlation coefficient.
c. Interpret the correlation coefficient. Does it surprise you that the correlation coefficient is negative?

Testing the Significance of the Correlation Coefficient

Recall that the sales manager of North American Copier Sales found the correlation between the number of sales calls and the number of copiers sold was 0.865. This indicated a strong positive association between the two variables. However, only 15 salespeople were sampled. Could it be that the correlation in the population is actually 0? This would mean the correlation of 0.865 was due to chance, or sampling error. The population in this example is all the salespeople employed by the firm.

Resolving this dilemma requires a test to answer the question: Could there be zero correlation in the population from which the sample was selected? To put it another way, did the computed r come from a population of paired observations with zero correlation? To continue our convention of allowing Greek letters to represent a population parameter, we will let ρ represent the correlation in the population. It is pronounced "rho."

We will continue with the illustration involving sales calls and copiers sold. We employ the same hypothesis testing steps described in Chapter 10. The null hypothesis and the alternate hypothesis are:

H_0: $\rho = 0$ (The correlation in the population is zero.)
H_1: $\rho \neq 0$ (The correlation in the population is different from zero.)

This is a two-tailed test. The null hypothesis can be rejected with either large or small sample values of the correlation coefficient.

The formula for t is:

t TEST FOR THE CORRELATION COEFFICIENT

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \text{ with } n-2 \text{ degrees of freedom} \qquad \textbf{[13–2]}$$

Using the .05 level of significance, the decision rule states that if the computed t falls in the area between plus 2.160 and minus 2.160, the null hypothesis is not rejected. To locate the critical value of 2.160, refer to Appendix B.5 for $df = n - 2 = 15 - 2 = 13$. See Chart 13–6.

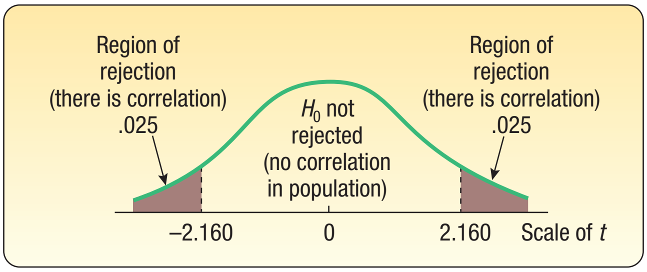


CHART 13–6 Decision Rule for Test of Hypothesis at .05 Significance Level and 13 *df*

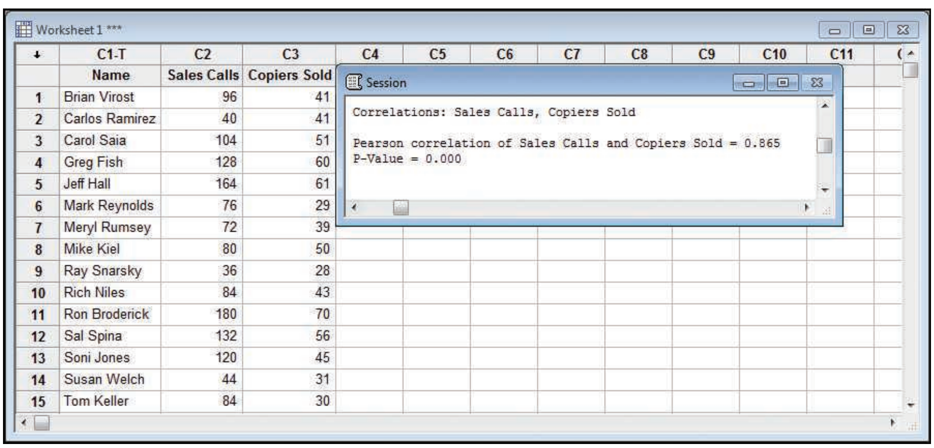
Applying formula (13–2) to the example regarding the number of sales calls and units sold:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{.865\sqrt{15-2}}{\sqrt{1-.865^2}} = 6.216$$

The computed t is in the rejection region. Thus, H_0 is rejected at the .05 significance level. Hence we conclude the correlation in the population is not zero. This indicates to the sales manager that there is correlation with respect to the number of sales calls made and the number of copiers sold in the population of salespeople.

We can also interpret the test of hypothesis in terms of p-values. A p-value is the likelihood of finding a value of the test statistic more extreme than the one computed, when H_0 is true. To determine the p-value, go to the t distribution in Appendix B.5 and find the row for 13 degrees of freedom. The value of the test statistic is 6.216, so in the row for 13 degrees of freedom and a two-tailed test, find the value closest to 6.216. For a two-tailed test at the 0.001 significance level, the critical value is 4.221. Because 6.216 is greater than 4.221, we conclude that the p-value is less than 0.001.

Both Minitab and Excel will report the correlation between two variables. In addition to the correlation, Minitab reports the p-value for the test of hypothesis that the correlation in the population between the two variables is 0. The Minitab output follows.



	C1-T Name	C2 Sales Calls	C3 Copiers Sold
1	Brian Virost	96	41
2	Carlos Ramirez	40	41
3	Carol Saia	104	51
4	Greg Fish	128	60
5	Jeff Hall	164	61
6	Mark Reynolds	76	29
7	Meryl Rumsey	72	39
8	Mike Kiel	80	50
9	Ray Snarsky	36	28
10	Rich Niles	84	43
11	Ron Broderick	180	70
12	Sal Spina	132	56
13	Soni Jones	120	45
14	Susan Welch	44	31
15	Tom Keller	84	30

```
Correlations: Sales Calls, Copiers Sold

Pearson correlation of Sales Calls and Copiers Sold = 0.865
P-Value = 0.000
```

EXAMPLE

In the Applewood Auto Group example on page 444, we found that the correlation coefficient between the profit on the sale of a vehicle by the Applewood Auto Group and the age of the person that purchased the vehicle was 0.262. The sign of the correlation coefficient was positive, so we concluded there was a direct relationship between the two variables. However, because the value of the correlation coefficient was small—that is, near zero—we concluded that an advertising campaign directed toward the older buyers was not warranted. We can test our conclusion by conducting a hypothesis test that the correlation coefficient is greater than zero using the .05 significance level.

SOLUTION

To test the hypothesis, we need to clarify the sample and population issues. Let's assume that the data collected on the 180 vehicles sold by the Applewood Group is a sample from the population of *all* vehicles sold over many years by the Applewood Auto Group. The Greek letter ρ is the correlation coefficient in the population and r the correlation coefficient in the sample.

Our next step is to set up the null hypothesis and the alternate hypothesis. We test the null hypothesis that the correlation coefficient is equal to or less than zero. The alternate hypothesis is that there is positive correlation between the two variables.

H_0: $\rho \leq 0$ (The correlation in the population is negative or equal to zero.)
H_1: $\rho > 0$ (The correlation in the population is positive.)

This is a one-tailed test because we are interested in confirming a positive association between the variables. The test statistic follows the t distribution with $n - 2$ degrees of freedom, so the degrees of freedom are $180 - 2 = 178$. However, the value for 178 degrees of freedom is not in Appendix B.5. The closest value is 180, so we will use that value. Our decision rule is to reject the null hypothesis if the computed value of the test statistic is greater than 1.653.

We use formula (13–2) to find the value of the test statistic.

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.262\sqrt{180-2}}{\sqrt{1-0.262^2}} = 3.622$$

Comparing the value of our test statistic of 3.622 to the critical value of 1.653, we reject the null hypothesis. We conclude that the sample correlation coefficient of 0.262 is too large to have come from a population with no correlation. To put our results another way, there is a positive correlation between profits and age in the population.

This result is confusing and seems contradictory. On one hand, we observed that the correlation coefficient did not indicate a very strong relationship and that the Applewood Auto Group marketing department should not use this information for its promotion and advertising decisions. On the other hand, the hypothesis test indicated that the correlation coefficient is not equal to zero and that a positive relationship between age and profit exists. How can this be? We must be very careful about the application of the hypothesis test results. The hypothesis test shows a statistically significant result. However, this result does not necessarily support a practical decision to start a new marketing and promotion campaign to older purchasers. In fact, the relatively low correlation coefficient is an indication that the outcome of a new marketing and promotion campaign to older potential purchasers is, at best, uncertain.

SELF-REVIEW 13–2

A sample of 25 mayoral campaigns in medium-sized cities with populations between 50,000 and 250,000 showed that the correlation between the percent of the vote received and the amount spent on the campaign by the candidate was .43. At the .05 significance level, is there a positive association between the variables?

EXERCISES

7. The following hypotheses are given.

$$H_0: \rho \le 0$$
$$H_1: \rho > 0$$

A random sample of 12 paired observations indicated a correlation of .32. Can we conclude that the correlation in the population is greater than zero? Use the .05 significance level.

8. The following hypotheses are given.

$$H_0: \rho \ge 0$$
$$H_1: \rho < 0$$

A random sample of 15 paired observations has a correlation of −.46. Can we conclude that the correlation in the population is less than zero? Use the .05 significance level.

9. Pennsylvania Refining Company is studying the relationship between the pump price of gasoline and the number of gallons sold. For a sample of 20 stations last Tuesday, the correlation was .78. At the .01 significance level, is the correlation in the population greater than zero?

10. A study of 20 worldwide financial institutions showed the correlation between their assets and pretax profit to be .86. At the .05 significance level, can we conclude that there is positive correlation in the population?

11. The Airline Passenger Association studied the relationship between the number of passengers on a particular flight and the cost of the flight. It seems logical that more passengers on the flight will result in more weight and more luggage, which in turn will result in higher fuel costs. For a sample of 15 flights, the correlation between the number of passengers and total fuel cost was .667. Is it reasonable to conclude that there is positive association in the population between the two variables? Use the .01 significance level.

12. **FILE** The Student Government Association at Middle Carolina University wanted to demonstrate the relationship between the number of beers a student drinks and his or her blood alcohol content (BAC). A random sample of 18 students participated in a study in which each participating student was randomly assigned a number of 12-ounce cans of beer to drink. Thirty minutes after they consumed their assigned number of beers, a member of the local sheriff's office measured their blood alcohol content. The sample information is reported below.

Student	Beers	BAC	Student	Beers	BAC
Charles	6	0.10	Jaime	3	0.07
Ellis	7	0.09	Shannon	3	0.05
Harriet	7	0.09	Nellie	7	0.08
Marlene	4	0.10	Jeanne	1	0.04
Tara	5	0.10	Michele	4	0.07
Kerry	3	0.07	Seth	2	0.06
Vera	3	0.10	Gilberto	7	0.12
Pat	6	0.12	Lillian	2	0.05
Marjorie	6	0.09	Becky	1	0.02

Use a statistical software package to answer the following questions.

a. Develop a scatter diagram for the number of beers consumed and BAC. Comment on the relationship. Does it appear to be strong or weak? Does it appear to be positive or inverse?

b. Determine the correlation coefficient.

c. At the .01 significance level, is it reasonable to conclude that there is a positive relationship in the population between the number of beers consumed and the BAC? What is the *p*-value?

LO13-3

Apply regression analysis to estimate the linear relationship between two variables.

STATISTICS IN ACTION

In finance, investors are interested in the trade-off between returns and risk. One technique to quantify risk is a regression analysis of a company's stock price (dependent variable) and an average measure of the stock market (independent variable). Often the Standard and Poor's (S&P) 500 Index is used to estimate the market. The regression coefficient, called beta in finance, shows the change in a company's stock price for a one-unit change in the S&P Index. For example, if a stock has a beta of 1.5, then when the S&P index increases by 1%, the stock price will increase by 1.5%. The opposite is also true. If the S&P decreases by 1%, the stock price will decrease by 1.5%. If the beta is 1.0, then a 1% change in the index should show a 1% change in a stock price. If the beta is less than 1.0, then a 1% change in the index shows less than a 1% change in the stock price.

REGRESSION ANALYSIS

In the previous sections of this chapter, we evaluated the direction and the significance of the linear relationship between two variables by finding the correlation coefficient. Regression analysis is another method to examine a linear relationship between two variables. This analysis uses the basic concepts of correlation but provides much more information by expressing the linear relationship between two variables in the form of an equation. Using this equation, we will be able to estimate the value of the dependent variable *Y* based on a selected value of the independent variable *X*. The technique used to develop the equation and provide the estimates is called **regression analysis.**

© Image Source/Getty Images RF

In Table 13–1, we reported the number of sales calls and the number of units sold for a sample of 15 sales representatives employed by North American Copier Sales. Chart 13–1 portrayed this information in a scatter diagram. Recall that we tested the significance of the correlation coefficient ($r = 0.865$) and concluded that a significant relationship exists between the two variables. Now we want to develop a linear equation that expresses the relationship between the number of sales calls, the independent variable, and the number of units sold, the dependent variable. The equation for the line used to estimate *Y* on the basis of *X* is referred to as the **regression equation.**

REGRESSION EQUATION An equation that expresses the linear relationship between two variables.

Least Squares Principle

In regression analysis, our objective is to use the data to position a line that best represents the relationship between the two variables. Our first approach is to use a scatter diagram to visually position the line.

The scatter diagram in Chart 13–1 is reproduced in Chart 13–7, with a line drawn with a ruler through the dots to illustrate that a line would probably fit the data. However, the line drawn using a straight edge has one disadvantage: Its position is based in part on the judgment of the person drawing the line. The hand-drawn lines in Chart 13–8 represent the judgments of four people. All the lines except line *A* seem to be reasonable. That is, each line is centered among the graphed data. However, each would result in a different estimate of units sold for a particular number of sales calls.

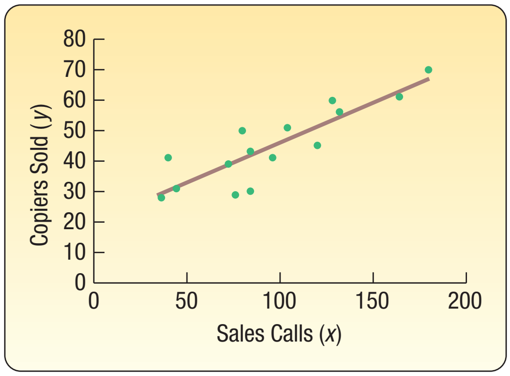


CHART 13–7 Sales Calls and Copiers Sold for 15 Sales Representatives

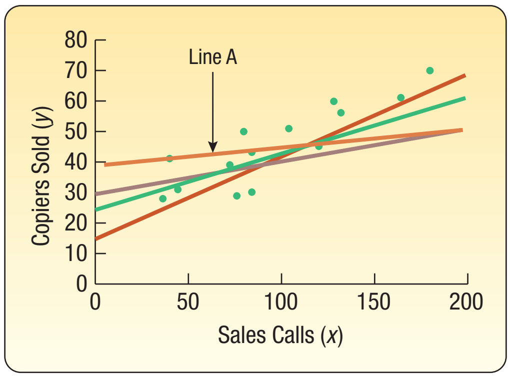


CHART 13–8 Four Lines Superimposed on the Scatter Diagram

We would prefer a method that results in a single, best regression line. This method is called the least squares principle. It gives what is commonly referred to as the "best-fitting" line.

LEAST SQUARES PRINCIPLE
A mathematical procedure that uses the data to position a line with the objective of minimizing the sum of the squares of the vertical distances between the actual y values and the predicted values of y.

To illustrate this concept, the same data are plotted in the three charts that follow. The dots are the actual values of y, and the asterisks are the predicted values of y for a given value of x. The regression line in Chart 13–9 was determined using the least squares method. It is the best-fitting line because the sum of the squares of the vertical deviations about it is at a minimum. The first plot ($x = 3$, $y = 8$) deviates by 2 from the line, found by $10 - 8$. The deviation squared is 4. The squared deviation for the plot $x = 4$, $y = 18$ is 16. The squared deviation for the plot $x = 5$, $y = 16$ is 4. The sum of the squared deviations is 24, found by $4 + 16 + 4$.

Assume that the lines in Charts 13–10 and 13–11 were drawn with a straight edge. The sum of the squared vertical deviations in Chart 13–10 is 44. For Chart 13–11, it is 132. Both sums are greater than the sum for the line in Chart 13–9, found by using the least squares method.

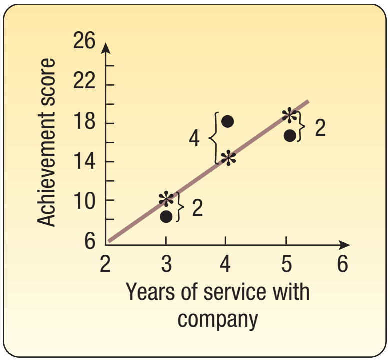


CHART 13–9 The Least Squares Line

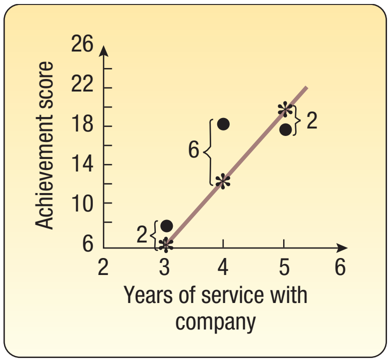


CHART 13–10 Line Drawn with a Straight Edge

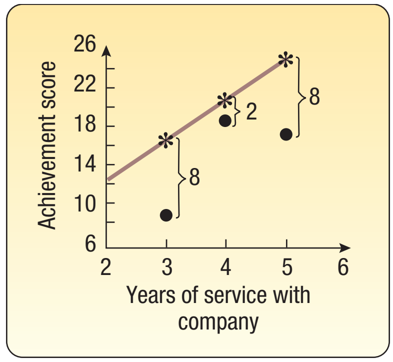


CHART 13–11 Different Line Drawn with a Straight Edge

The equation of a line has the form

GENERAL FORM OF LINEAR REGRESSION EQUATION

$$\hat{y} = a + bx \quad \text{[13–3]}$$

where:

- $\hat{y}$, read y hat, is the estimated value of the y variable for a selected x value.
- a is the y-intercept. It is the estimated value of Y when $x = 0$. Another way to put it is: a is the estimated value of y where the regression line crosses the Y-axis when x is zero.
- b is the slope of the line, or the average change in $\hat{y}$ for each change of one unit (either increase or decrease) in the independent variable x.
- x is any value of the independent variable that is selected.

The general form of the linear regression equation is exactly the same form as the equation of any line. a is the Y intercept and b is the slope. The purpose of regression analysis is to calculate the values of a and b to develop a linear equation that best fits the data.

The formulas for a and b are:

SLOPE OF THE REGRESSION LINE

$$b = r\left(\frac{s_y}{s_x}\right) \quad \text{[13–4]}$$

where:

- r is the correlation coefficient.
- s_y is the standard deviation of y (the dependent variable).
- s_x is the standard deviation of x (the independent variable).

***Y*-INTERCEPT**

$$a = \bar{y} - b\bar{x} \quad \text{[13–5]}$$

where:

- $\bar{y}$ is the mean of y (the dependent variable).
- $\bar{x}$ is the mean of x (the independent variable).

EXAMPLE

Recall the example involving North American Copier Sales. The sales manager gathered information on the number of sales calls made and the number of copiers sold for a random sample of 15 sales representatives. As a part of her presentation at the upcoming sales meeting, Ms. Bancer, the sales manager, would like to offer specific information about the relationship between the number of sales calls and the number of copiers sold. Use the least squares method to determine a linear equation to express the relationship between the two variables. What is the expected number of copiers sold by a representative who made 100 calls?

SOLUTION

The first step in determining the regression equation is to find the slope of the least squares regression line. That is, we need the value of b. In the previous section on page 443, we determined the correlation coefficient r (0.865). In the Excel output on page 443, we determined the standard deviation of the independent variable x (42.76) and the standard deviation of the dependent variable y (12.89). The values are inserted in formula (13–4).

$$b = r\left(\frac{s_y}{s_x}\right) = .865\left(\frac{12.89}{42.76}\right) = 0.2608$$

Next, we need to find the value of a. To do this, we use the value for b that we just calculated as well as the means for the number of sales calls and the number of copiers sold. These means are also available in the Excel worksheet on page 443. From formula (13–5):

$$a = \overline{y} - b\overline{x} = 45 - .2608(96) = 19.9632$$

Thus, the regression equation is

$$\hat{y} = 19.9632 + 0.2608x.$$

So if a salesperson makes 100 calls, he or she can expect to sell 46.0432 copiers, found by

$$\hat{y} = 19.9632 + 0.2608x = 19.9632 + 0.2608(100) = 46.0432$$

The b value of .2608 indicates that for each additional sales call, the sales representative can expect to increase the number of copiers sold by about 0.2608. To put it another way, 20 additional sales calls in a month will result in about five more copiers being sold, found by 0.2608 (20) = 5.216.

The a value of 19.9632 is the point where the equation crosses the Y-axis. A literal translation is that if no sales calls are made, that is $x = 0$, 19.9632 copiers will be sold. Note that $x = 0$ is outside the range of values included in the sample and, therefore, should not be used to estimate the number of copiers sold. The sales calls ranged from 36 to 180, so estimates should be limited to that range.

Drawing the Regression Line

The least squares equation $\hat{y} = 19.9632 + 0.2608x$ can be drawn on the scatter diagram. The fifth sales representative in the sample is Jeff Hall. He made 164 calls. His estimated number of copiers sold is $\hat{y} = 19.9632 + 0.2608(164) = 62.7344$. The plot $x = 164$ and $\hat{y} = 62.7344$ is located by moving to 164 on the X-axis and then going vertically to 62.7344. The other points on the regression equation can be determined

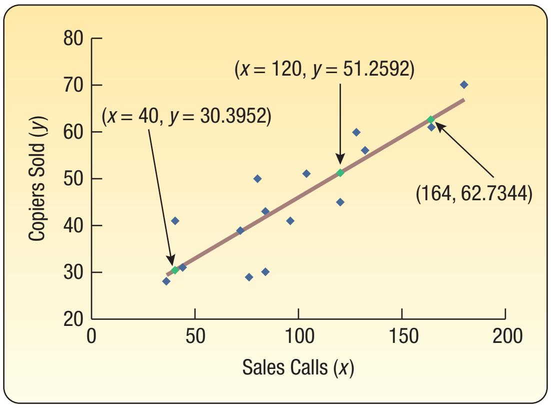


CHART 13–12 The Line of Regression Drawn on the Scatter Diagram

by substituting a particular value of x into the regression equation and calculating $\hat{y}$. All the points are connected to give the line. See Chart 13–12.

Sales Representative	Sales Calls (x)	Copiers Sold (y)	Estimated Sales ($\hat{y}$)
Brian Virost	96	41	45.0000
Carlos Ramirez	40	41	30.3952
Carol Saia	104	51	47.0864
Greg Fish	128	60	53.3456
Jeff Hall	164	61	62.7344
Mark Reynolds	76	29	39.7840
Meryl Rumsey	72	39	38.7408
Mike Kiel	80	50	40.8272
Ray Snarsky	36	28	29.3520
Rich Niles	84	43	41.8704
Ron Broderick	180	70	66.9072
Sal Spina	132	56	54.3888
Soni Jones	120	45	51.2592
Susan Welch	44	31	31.4384
Tom Keller	84	30	41.8704

The least squares regression line has some interesting and unique features. First, it will always pass through the point ($\bar{x}$, $\bar{y}$). To show this is true, we can use the mean number of sales calls to predict the number of copiers sold. In this example, the mean number of sales calls is 96, found by $\bar{x} = 1440/15$. The mean number of copiers sold is 45.0, found by $\bar{y} = 675/15$. If we let $x = 96$ and then use the regression equation to find the estimated value for $\hat{y}$ the result is:

$$\hat{y} = 19.9632 + 0.2608(96) = 45$$

The estimated number of copiers sold is exactly equal to the mean number of copiers sold. This simple example shows the regression line will pass through the point represented by the two means. In this case, the regression equation will pass through the point $x = 96$ and $y = 45$.

Second, as we discussed earlier in this section, there is no other line through the data where the sum of the squared deviations is smaller. To put it another way, the term $\Sigma(y - \hat{y})^2$ is smaller for the least squares regression equation than for any other equation. We use the Excel system to demonstrate this result in the following printout.

	A	B	C	D	E	F	G	H	I	J
1	**Sales Rep**	**Sales Calls (x)**	**Copiers Sold (y)**	**Estimated Sales**	**$(y-\hat{y})$**	**$(y-\hat{y})^2$**	**y^***	**$(y-y^*)^2$**	**y^{**}**	**$(y-y^{**})^2$**
2	Brian Virost	96	41	45.0000	–4.0000	16.0000	44.4000	11.5600	41.6000	0.3600
3	Carlos Ramirez	40	41	30.3952	10.6048	112.4618	29.0000	144.0000	29.0000	144.0000
4	Carol Saia	104	51	47.0864	3.9136	15.3163	46.6000	19.3600	43.4000	57.7600
5	Greg Fish	128	60	53.3456	6.6544	44.2810	53.2000	46.2400	48.8000	125.4400
6	Jeff Hall	164	61	62.7344	–1.7344	3.0081	63.1000	4.4100	56.9000	16.8100
7	Mark Reynolds	76	29	39.7840	–10.7840	116.2947	38.9000	98.0100	37.1000	65.6100
8	Meryl Rumsey	72	39	38.7408	0.2592	0.0672	37.8000	1.4400	36.2000	7.8400
9	Mike Kiel	80	50	40.8272	9.1728	84.1403	40.0000	100.0000	38.0000	144.0000
10	Ray Snarsky	36	28	29.3520	–1.3520	1.8279	27.9000	0.0100	28.1000	0.0100
11	Rich Niles	84	43	41.8704	1.1296	1.2760	41.1000	3.6100	38.9000	16.8100
12	Ron Broderick	180	70	66.9072	3.0928	9.5654	67.5000	6.2500	60.5000	90.2500
13	Sal Spina	132	56	54.3888	1.6112	2.5960	54.3000	2.8900	49.7000	39.6900
14	Soni Jones	120	45	51.2592	–6.2592	39.1776	51.0000	36.0000	47.0000	4.0000
15	Susan Welch	44	31	31.4384	–0.4384	0.1922	30.1000	0.8100	29.9000	1.2100
16	Tom Keller	84	30	41.8704	–11.8704	140.9064	41.1000	123.2100	38.9000	79.2100
17	**Total**				**0.0000**	**587.1108**		**597.8000**		**793.0000**

In columns A, B, and C in the Excel spreadsheet above, we duplicated the sample information on sales and copiers sold from Table 13–1. In column D, we provide the estimated sales values, the $\hat{y}$ values, as calculated above.

In column E, we calculate the **residuals,** or the error values. This is the difference between the actual values and the predicted values. That is, column E is $(y - \hat{y})$. For Soni Jones,

$$\hat{y} = 19, 9632 + 0.2608(120) = 51.2592$$

Her actual value is 45. So the residual, or error of estimate, is

$$(y - \hat{y}) = (45 - 51.2592) = -6.2592$$

This value reflects the amount the predicted value of sales is "off" from the actual sales value.

Next, in Column F, we square the residuals for each of the sales representatives and total the result. The total is 587.111.

$$\Sigma(y - \hat{y})^2 = 16.0000 + 112.4618 + \cdots + 140.9064 = 587.1108$$

This is the sum of the squared differences or the least squares value. There is no other line through these 15 data points where the sum of the squared differences is smaller.

We can demonstrate the least squares criterion by choosing two arbitrary equations that are close to the least squares equation and determining the sum of the squared differences for these equations. In column G, we use the equation $y^* = 18 + 0.275x$ to find the predicted value. Notice this equation is very similar to the least squares equation. In column H, we determine the residuals and square these residuals. For the first sales representative, Brian Virost,

$$y^* = 18 + 0.275(96) = 44.4$$

$$(y - y^*)^2 = (41 - 44.4)^2 = 11.56$$

This procedure is continued for the other 14 sales representatives and the squared residuals totaled. The result is 597.8. This is a larger value (597.8 is more than 587.1108) than the residuals for the least squares line.

In columns I and J on the output, we repeat the above process for yet another equation $y^{**} = 20 + 0.225x$. Again, this equation is similar to the least squares equation. The details for Brian Virost are:

$$y^{**} = 20 + 0.225x = 20 + 0.225(96) = 41.6$$

$$(y - y^{**})^2 = (41 - 41.6)^2 = 0.36$$

This procedure is continued for the other 14 sales representatives and the residuals totaled. The result is 793, which is also larger than the least squares values.

What have we shown with the example? The sum of the squared residuals $[\Sigma(y - \hat{y})^2]$ for the least squares equation is smaller than for other selected lines. The bottom line is you will not be able to find a line passing through these data points where the sum of the squared residuals is smaller.

SELF-REVIEW 13–3

Refer to Self-Review 13–1, where the owner of Haverty's Furniture Company was studying the relationship between sales and the amount spent on advertising. The advertising expense and sales revenue, both in millions of dollars, for the last 4 months are repeated below.

Month	Advertising Expense ($ million)	Sales Revenue ($ million)
July	2	7
August	1	3
September	3	8
October	4	10

(a) Determine the regression equation.
(b) Interpret the values of *a* and *b*.
(c) Estimate sales when $3 million is spent on advertising.

EXERCISES

13. **FILE** The following sample of observations was randomly selected.

x:	4	5	3	6	10
y:	4	6	5	7	7

a. Determine the regression equation.
b. Determine the value of when *x* is 7.

14. **FILE** The following sample of observations was randomly selected.

x:	5	3	6	3	4	4	6	8
y:	13	15	7	12	13	11	9	5

a. Determine the regression equation.
b. Determine the value of $\hat{y}$ when *x* is 7.

15. **FILE** Bradford Electric Illuminating Company is studying the relationship between kilowatt-hours (thousands) used and the number of rooms in a private single-family residence. A random sample of 10 homes yielded the following.

Number of Rooms	Kilowatt-Hours (thousands)	Number of Rooms	Kilowatt-Hours (thousands)
12	9	8	6
9	7	10	8
14	10	10	10
6	5	5	4
10	8	7	7

a. Determine the regression equation.
b. Determine the number of kilowatt-hours, in thousands, for a six-room house.

16. **FILE** Mr. James McWhinney, president of Daniel-James Financial Services, believes there is a relationship between the number of client contacts and the dollar amount of sales. To document this assertion, Mr. McWhinney gathered the following sample information. The x column indicates the number of client contacts last month and the y column shows the value of sales ($ thousands) last month for each client sampled.

Number of Contacts, x	Sales ($ thousands), y	Number of Contacts, x	Sales ($ thousands), y
14	24	23	30
12	14	48	90
20	28	50	85
16	30	55	120
46	80	50	110

a. Determine the regression equation.
b. Determine the estimated sales if 40 contacts are made.

17. **FILE** A recent article in *Bloomberg Businessweek* listed the "Best Small Companies." We are interested in the current results of the companies' sales and earnings. A random sample of 12 companies was selected and the sales and earnings, in millions of dollars, are reported below.

Company	Sales ($ millions)	Earnings ($ millions)	Company	Sales ($ millions)	Earnings ($ millions)
Papa John's International	$89.2	$4.9	Checkmate Electronics	$17.5	$ 2.6
Applied Innovation	18.6	4.4	Royal Grip	11.9	1.7
Integracare	18.2	1.3	M-Wave	19.6	3.5
Wall Data	71.7	8.0	Serving-N-Slide	51.2	8.2
Davidson & Associates	58.6	6.6	Daig	28.6	6.0
Chico's FAS	46.8	4.1	Cobra Golf	69.2	12.8

Let sales be the independent variable and earnings be the dependent variable.
a. Draw a scatter diagram.
b. Compute the correlation coefficient.
c. Determine the regression equation.
d. For a small company with $50.0 million in sales, estimate the earnings.

18. **FILE** We are studying mutual bond funds for the purpose of investing in several funds. For this particular study, we want to focus on the assets of a fund and its five-year performance. The question is: Can the five-year rate of return be estimated based on the assets of the fund? Nine mutual funds were selected at random, and their assets and rates of return are shown below.

Fund	Assets ($ millions)	Return (%)	Fund	Assets ($ millions)	Return (%)
AARP High Quality Bond	$622.2	10.8	MFS Bond A	$494.5	11.6
Babson Bond L	160.4	11.3	Nichols Income	158.3	9.5
Compass Capital Fixed Income	275.7	11.4	T. Rowe Price Short-term	681.0	8.2
Galaxy Bond Retail	433.2	9.1	Thompson Income B	241.3	6.8
Keystone Custodian B-1	437.9	9.2			

a. Draw a scatter diagram.
b. Compute the correlation coefficient.
c. Write a brief report of your findings for parts (a) and (b).
d. Determine the regression equation. Use assets as the independent variable.
e. For a fund with $400.0 million in sales, determine the five-year rate of return (in percent).

19. FILE Refer to Exercise 5. Assume the dependent variable is number of crimes.
a. Determine the regression equation.
b. Estimate the number of crimes for a city with 20 police officers.
c. Interpret the regression equation.

20. FILE Refer to Exercise 6.
a. Determine the regression equation.
b. Estimate the selling price of a 10-year-old car.
c. Interpret the regression equation.

LO13-4
Evaluate the significance of the slope of the regression equation.

TESTING THE SIGNIFICANCE OF THE SLOPE

In the prior section, we showed how to find the equation of the regression line that best fits the data. The method for finding the equation is based on the *least squares principle*. The purpose of the regression equation is to quantify a linear relationship between two variables.

The next step is to analyze the regression equation by conducting a test of hypothesis to see if the slope of the regression line is different from zero. Why is this important? If we can show that the slope of the line in the population is different from zero, then we can conclude that using the regression equation adds to our ability to predict or forecast the dependent variable based on the independent variable. If we cannot demonstrate that this slope is different from zero, then we conclude there is no merit to using the independent variable as a predictor. To put it another way, if we cannot show the slope of the line is different from zero, we might as well use the mean of the dependent variable as a predictor, rather than use the regression equation.

Following from the hypothesis-testing procedure in Chapter 10, the null and alternative hypotheses are:

$$H_0: \beta = 0$$
$$H_1: \beta \neq 0$$

We use β (the Greek letter beta) to represent the population slope for the regression equation. This is consistent with our policy to identify population parameters by Greek letters. We assumed the information regarding North American Copier Sales, Table 13–2, is a sample. Be careful here. Remember, this is a single sample of salespeople, but when we selected a particular salesperson we identified two pieces of information: how many customers they called on and how many copiers they sold.

We identified the slope value as b. So b is our computed slope based on a sample and is an estimate of the population's slope, identified as β. The null hypothesis is that the slope of the regression equation in the population is zero. If this is the case, the regression line is horizontal and there is no relationship between the independent variable, X, and the dependent variable, Y. In other words, the value of the dependent variable is the same for any value of the independent variable and does not offer us any help in estimating the value of the dependent variable.

What if the null hypothesis is rejected? If the null hypothesis is rejected and the alternate hypothesis accepted, this indicates that the slope of the regression line for the population is not equal to zero. To put it another way, a significant relationship exists between the two variables. Knowing the value of the independent variable allows us to estimate the value of the dependent variable.

Before we test the hypothesis, we use statistical software to determine the needed regression statistics. We continue to use the North American Copier Sales data from Table 13–2 and use Excel to perform the necessary calculations. The following spreadsheet shows three tables to the right of the sample data.

	A	B	C	D	E	F	G	H	I	J
1	**Sales Representative**	**Sales calls (x)**	**Copiers Sold (y)**		SUMMARY OUTPUT					
2	Brian Virost	96	41							
3	Carlos Ramirez	40	41		*Regression Statistics*					
4	Carol Saia	104	51		Multiple R	0.865				
5	Greg Fish	128	60		R Square	0.748				
6	Jeff Hall	164	61		Adjusted R Square	0.728				
7	Mark Reynolds	76	29		Standard Error	6.720				
8	Meryl Rumsey	72	39		Observations	15				
9	Mike Kiel	80	50							
10	Ray Snarsky	36	28		ANOVA					
11	Rich Niles	84	43			*df*	*SS*	*MS*	*F*	*Significance F*
12	Ron Broderick	180	70		Regression	1	1738.89	1738.89	38.5031	3.19277E-05
13	Sal Spina	132	56		Residual	13	587.11	45.1623		
14	Sani Jones	120	45		Total	14	2326			
15	Susan Welch	44	31							
16	Tom Keller	84	30			*Coefficients*	*Standard Error*	*t Stat*	*P-value*	
17					Intercept	19.9800	4.389675533	4.55159	0.00054	
18					Sales calls (x)	0.2606	0.042001817	6.20509	3.2E-05	

1. Starting on the top are the *Regression Statistics*. We will use this information later in the chapter, but notice that the "Multiple R" value is familiar. It is .865, which is the correlation coefficient we calculated using formula (13–1).
2. Next is an ANOVA table. This is a useful table for summarizing regression information. We will refer to it later in this chapter and use it extensively in the next chapter when we study multiple regression.
3. At the bottom, highlighted in blue, is the information needed to conduct our test of hypothesis regarding the slope of the line. It includes the value of the slope, which is 0.2606, and the intercept, which is 19.98. (Note that these values for the slope and the intercept are slightly different from those computed in the Example/Solution on page 454. These small differences are due to rounding.) In the column to the right of the regression coefficient is a column labeled "Standard Error." This is a value similar to the standard error of the mean. Recall that the standard error of the mean reports the variation in the sample means. In a similar fashion, these standard errors report the possible variation in slope and intercept values. The standard error of the slope coefficient is 0.0420.

To test the null hypothesis, we use the *t*-distribution with $(n - 2)$ and the following formula.

TEST FOR THE SLOPE $$t = \frac{b - 0}{s_b} \quad \text{with } n - 2 \text{ degrees of freedom} \qquad \textbf{[13–6]}$$

where:

b is the estimate of the regression line's slope calculated from the sample information.

s_b is the standard error of the slope estimate, also determined from sample information.

Our first step is to set the null and the alternative hypotheses. They are:

$$H_0: \beta \leq 0$$

$$H_1: \beta > 0$$

Notice that we have a one-tailed test. If we do not reject the null hypothesis, we conclude that the slope of the regression line in the population could be zero. This means the independent variable is of no value in improving our estimate of the dependent variable. In our case, this means that knowing the number of sales calls made by a representative does not help us predict the sales.

If we reject the null hypothesis and accept the alternative, we conclude the slope of the line is greater than zero. Hence, the independent variable is an aid in predicting the dependent variable. Thus, if we know the number of sales calls made by a salesperson, we can predict or forecast their sales. We also know, because we have demonstrated that the slope of the line is greater than zero—that is, positive—that more sales calls will result in the sale of more copiers.

The *t*-distribution is the test statistic; there are 13 degrees of freedom, found by $n - 2 = 15 - 2$. We use the .05 significance level. From Appendix B.5, the critical value is 1.771. Our decision rule is to reject the null hypothesis if the value computed from formula (13–6) is greater than 1.771. We apply formula (13–6) to find t.

$$t = \frac{b - 0}{s_b} = \frac{0.2606 - 0}{0.042} = 6.205$$

The computed value of 6.205 exceeds our critical value of 1.771, so we reject the null hypothesis and accept the alternative hypothesis. We conclude that the slope of the line is greater than zero. The independent variable, number of sales calls, is useful in estimating copier sales.

The table also provides us information on the *p*-value of this test. This cell is highlighted in purple. So we could select a significance level, say .05, and compare that value with the *p*-value. In this case, the calculated *p*-value in the table is reported in exponential notation and is equal to 0.0000319, so our decision is to reject the null hypothesis. An important caution is that the *p*-values reported in the statistical software are usually for a two-*tailed test.*

Before moving on, here is an interesting note. Observe that on page 448, when we conducted a test of hypothesis regarding the correlation coefficient for these same data using formula (13–2), we obtained the same value of the *t*-statistic, $t = 6.205$. Actually, when comparing the results of simple linear regression and correlation analysis, the two tests are equivalent and will always yield exactly the same values of t and the same *p*-values.

SELF-REVIEW 13–4

Refer to Self-Review 13–1, where the owner of Haverty's Furniture Company studied the relationship between the amount spent on advertising in a month and sales revenue for that month. The amount of sales is the dependent variable and advertising expense, the independent variable. The regression equation in that study was $\hat{y} = 1.5 + 2.2x$ for a sample of 5 months. Conduct a test of hypothesis to show there is a positive relationship between advertising and sales. From statistical software, the standard error of the regression coefficient is 0.42. Use the .05 significance level.

EXERCISES

21. **FILE** Refer to Exercise 5. The regression equation is $\hat{y} = 29.29 - 0.96x$, the sample size is 8, and the standard error of the slope is 0.22. Use the .05 significance level. Can we conclude that the slope of the regression line is less than zero?

22. **FILE** Refer to Exercise 6. The regression equation is $\hat{y} = 11.18 - 0.49x$, the sample size is 12, and the standard error of the slope is 0.23. Use the .05 significance level. Can we conclude that the slope of the regression line is less than zero?

23. **FILE** Refer to Exercise 17. The regression equation is $\hat{y} = 1.85 + .08x$, the sample size is 12, and the standard error of the slope is 0.03. Use the .05 significance level. Can we conclude that the slope of the regression line is *different from zero*?

24. **FILE** Refer to Exercise 18. The regression equation is $\hat{y} = 9.9198 - 0.00039x$, the sample size is 9, and the standard error of the slope is 0.0032. Use the .05 significance level. Can we conclude that the slope of the regression line is less than zero?

LO13-5
Evaluate a regression equation's ability to predict using the standard estimate of the error and the coefficient of determination.

EVALUATING A REGRESSION EQUATION'S ABILITY TO PREDICT

The Standard Error of Estimate

The results of the regression analysis for North American Copier Sales show a significant relationship between number of sales calls and the number of sales made. By substituting the names of the variables into the equation, it can be written as:

$$\text{Number of copiers sold} = 19.9632 + 0.2608\ (\text{Number of sales calls})$$

The equation can be used to estimate the number of copiers sold for any given "number of sales calls" within the range of the data. For example, if the number of sales calls is 84, then we can predict the number of copiers sold. It is 41.8704, found by 19.9632 + 0.2608(84). However, the data show two sales representatives with 84 sales calls and 30 and 43 copiers sold. So, is the regression equation a good predictor of "Number of copiers sold"?

Perfect prediction, which is finding the exact outcome, is practically impossible in almost all disciplines including economics and business. For example:

- A large electronics firm, with production facilities throughout the United States, has a stock option plan for employees. Suppose there is a relationship between the number of years employed and the number of shares owned. This relationship is likely because, as number of years of service increases, the number of shares an employee earns also increases. If we observe all employees with 20 years of service, they would most likely own different numbers of shares.
- A real estate developer in the southwest United States studied the relationship between the income of buyers and the size, in square feet, of the home they purchased. The developer's analysis shows that as the income of a purchaser increases, the size of the home purchased will also increase. However, all buyers with an income of $70,000 will not purchase a home of exactly the same size.

What is needed, then, is a measure that describes how precise the prediction of Y is based on X or, conversely, how inaccurate the estimate might be. This measure is called the **standard error of estimate.** The standard error of estimate is symbolized by $s_{y \cdot x}$. The subscript, $y \cdot x$, is interpreted as the standard error of y for a given value of x. It is the same concept as the standard deviation discussed in Chapter 3. The standard deviation measures the dispersion around the mean. The standard error of estimate measures the dispersion about the regression line for a given value of x.

STANDARD ERROR OF ESTIMATE A measure of the dispersion, or scatter, of the observed values around the line of regression for a given value of x.

The standard error of estimate is found using formula (13–7).

STANDARD ERROR OF ESTIMATE

$$s_{y \cdot x} = \sqrt{\frac{\Sigma(y - \hat{y})^2}{n - 2}} \qquad \textbf{[13–7]}$$

The calculation of the standard error of estimate requires the sum of the squared differences between each observed value of y and the predicted value of y, which is identified as $\hat{y}$ in the numerator. This calculation is illustrated in the following spreadsheet. See the highlighted cell in the lower right corner.

	A	B	C	D	E	F
1	**Sales Rep**	**Sales Calls (x)**	**Copiers Sold (y)**	**Estimated Sales**	**$(y-\hat{y})$**	**$(y-\hat{y})^2$**
2	Brian Virost	96	41	45.0000	–4.0000	16.0000
3	Carlos Ramirez	40	41	30.3952	10.6048	112.4618
4	Carol Saia	104	51	47.0864	3.9136	15.3163
5	Greg Fish	128	60	53.3456	6.6544	44.2810
6	Jeff Hall	164	61	62.7344	–1.7344	3.0081
7	Mark Reynolds	76	29	39.7840	–10.7840	116.2947
8	Meryl Rumsey	72	39	38.7408	0.2592	0.0672
9	Mike Kiel	80	50	40.8272	9.1728	84.1403
10	Ray Snarsky	36	28	29.3520	–1.3520	1.8279
11	Rich Niles	84	43	41.8704	1.1296	1.2760
12	Ron Broderick	180	70	66.9072	3.0928	9.5654
13	Sal Spina	132	56	54.3888	1.6112	2.5960
14	Soni Jones	120	45	51.2592	–6.2592	39.1776
15	Susan Welch	44	31	31.4384	–0.4384	0.1922
16	Tom Keller	84	30	41.8704	–11.8704	140.9064
17	**Total**				**0.0000**	**587.1108**

The calculation of the standard error of estimate is:

$$s_{y \cdot x} = \sqrt{\frac{\Sigma(y-\hat{y})^2}{n-2}} = \sqrt{\frac{587.1108}{15-2}} = 6.720$$

The standard error of estimate can be calculated using statistical software such as Excel. It is included in Excel's regression analysis on page 460 and highlighted in yellow. Its value is 6.720.

If the standard error of estimate is small, this indicates that the data are relatively close to the regression line and the regression equation can be used to predict *y* with little error. If the standard error of estimate is large, this indicates that the data are widely scattered around the regression line and the regression equation will not provide a precise estimate of *y*.

The Coefficient of Determination

Using the standard error of estimate provides a relative measure of a regression equation's ability to predict. We will use it to provide more specific information about a prediction in the next section. In this section, another statistic is explained that will provide a more interpretable measure of a regression equation's ability to predict. It is called the coefficient of determination, or *R*-square.

COEFFICIENT OF DETERMINATION
The proportion of the total variation in the dependent variable *Y* that is explained, or accounted for, by the variation in the independent variable *X*.

The coefficient of determination is easy to compute. It is the correlation coefficient squared. Therefore, the term *R*-square is also used. With the North American Copier Sales data, the correlation coefficient for the relationship between the number of copiers sold and the number of sales calls is 0.865. If we compute $(0.865)^2$, the coefficient of determination is 0.748. See the blue (Multiple R) and green (*R*-square) highlighted cells in the spreadsheet on page 460. To better interpret the coefficient of determination, convert it to a percentage. Hence, we say that 74.8% of the variation in the number of copiers sold is explained, or accounted for, by the variation in the number of sales calls.

How well can the regression equation predict number of copiers sold with number of sales calls made? If it were possible to make perfect predictions, the coefficient of determination would be 100%. That would mean that the independent variable, number of sales calls, explains or accounts for all the variation in the number of copiers sold. A coefficient of determination of 100% is associated with a correlation coefficient of +1.0 or −1.0. Refer to Chart 13–2, which shows that a perfect prediction is associated with a perfect linear relationship where all the data points form a perfect line in a scatter diagram. Our analysis shows that only 74.8% of the variation in copiers sold is explained by the number of sales calls. Clearly, these data do not form a perfect line. Instead, the data are scattered around the best-fitting, least squares regression line, and there will be error in the predictions. In the next section, the standard error of estimate is used to provide more specific information regarding the error associated with using the regression equation to make predictions.

SELF-REVIEW 13–5

Refer to Self-Review 13–1, where the owner of Haverty's Furniture Company studied the relationship between the amount spent on advertising in a month and sales revenue for that month. The amount of sales is the dependent variable and advertising expense is the independent variable.

(a) Determine the standard error of estimate.
(b) Determine the coefficient of determination.
(c) Interpret the coefficient of determination.

EXERCISES

(You may wish to use a statistical software package such as Excel, Minitab, or Megastat to assist in your calculations.)

25. Refer to Exercise 5. Determine the standard error of estimate and the coefficient of determination. Interpret the coefficient of determination.
26. Refer to Exercise 6. Determine the standard error of estimate and the coefficient of determination. Interpret the coefficient of determination.
27. Refer to Exercise 15. Determine the standard error of estimate and the coefficient of determination. Interpret the coefficient of determination.
28. Refer to Exercise 16. Determine the standard error of estimate and the coefficient of determination. Interpret the coefficient of determination.

Relationships among the Correlation Coefficient, the Coefficient of Determination, and the Standard Error of Estimate

In formula 13-7 shown on page 462, we described the standard error of estimate. Recall that it measures how close the actual values are to the regression line. When the standard error is small, it indicates that the two variables are closely related. In the calculation of the standard error, the key term is

$$\Sigma(y - \hat{y})^2$$

If the value of this term is small, then the standard error will also be small.

The correlation coefficient measures the strength of the linear association between two variables. When the points on the scatter diagram appear close to the line, we note that the correlation coefficient tends to be large. Therefore, the correlation coefficient and the standard error of the estimate are inversely related. As the strength of a linear relationship between two variables increases, the correlation coefficient increases and the standard error of the estimate decreases.

We also noted that the square of the correlation coefficient is the coefficient of determination. The coefficient of determination measures the percentage of the variation in *Y* that is explained by the variation in *X*.

A convenient vehicle for showing the relationship among these three measures is an ANOVA table. See the highlighted portion of the spreadsheet below. This table is similar to the analysis of variance table developed in Chapter 12. In that chapter, the total variation was divided into two components: variation due to the *treatments* and variation due to *random error.* The concept is similar in regression analysis. The total variation is divided into two components: (1) variation explained by the *regression* (explained by the independent variable) and (2) the *error,* or *residual.* This is the unexplained variation. These three sources of variance (total, regression, and residual) are identified in the first column of the spreadsheet ANOVA table. The column headed "*df*" refers to the degrees of freedom associated with each category. The total number of degrees of freedom is $n - 1$. The number of degrees of freedom in the regression is 1 because there is only one independent variable. The number of degrees of freedom associated with the error term is $n - 2$. The term SS located in the middle of the ANOVA table refers to the sum of squares. You should note that the total degrees of freedom are equal to the sum of the regression and residual (error) degrees of freedom, and the total sum of squares is equal to the sum of the regression and residual (error) sum of squares. This is true for any ANOVA table.

	A	B	C	D	E	F	G	H	I	J
1	**Sales Representive**	**Sales Calls (x)**	**Copiers Sold (y)**		**SUMMARY OUTPUT**					
2	Brian Virost	96	41		***Regression Statistics***					
3	Carlos Ramirez	40	41		Multiple R	0.865				
4	Carol Saia	104	51		R Square	0.748				
5	Greg Fish	128	60		Adjusted R Square	0.728				
6	Jeff Hall	164	61		Standard Error	6.720				
7	Mark Reynolds	76	29		Observations	15				
8	Meryl Rumsey	72	39							
9	Mike Kiel	80	50		**ANOVA**					
10	Ray Snarsky	36	28			**df**	**SS**	**MS**	**F**	**Significance F**
11	Rich Niles	84	43		Regression	1	1738.890	1738.890	38.503	0.000
12	Ron Broderick	180	70		Residual	13	587.110	45.162		
13	Sal Spina	132	56		Total	14	2326			
14	Sani Jones	120	45							
15	Susan Welch	44	31			***Coefficients***	***Standard Error***	***t Stat***	***P-value***	
16	Tom Keller	84	30		Intercept	19.980	4.390	4.552	0.001	
17					Sales Calls (x)	0.261	0.042	6.205	0.000	

The ANOVA sum of squares are:

$$\text{Regression Sum of Squares} = \text{SSR} = \Sigma(\hat{y} - \bar{y})^2 = 1738.89$$

$$\text{Residual or Error Sum of Squares} = \text{SSE} = \Sigma(y - \hat{y})^2 = 587.11$$

$$\text{Total Sum of Squares} = \text{SS Total} = \Sigma(y - \bar{y})^2 = 2326.0$$

Recall that the coefficient of determination is defined as the percentage of the total variation (SS Total) explained by the regression equation (SSR). Using the ANOVA table, the reported value of *R*-square can be validated.

COEFFICIENT OF DETERMINATION

$$r^2 = \frac{\text{SSR}}{\text{SS Total}} = 1 - \frac{\text{SSE}}{\text{SS Total}} \qquad \textbf{[13–8]}$$

Using the values from the ANOVA table, the coefficient of determination is 1738.89/2326.0 = 0.748. Therefore, the more variation of the dependent variable (SS Total) explained by the independent variable (SSR), the higher the coefficient of determination.

We can also express the coefficient of determination in terms of the error or residual variation:

$$r^2 = 1 - \frac{\text{SSE}}{\text{SS Total}} = 1 - \frac{587.11}{2326.0} = 1 - 0.252 = 0.748$$

As illustrated in formula 13–8, the coefficient of determination and the residual or error sum of squares are inversely related. The higher the unexplained or error variation as a percentage of the total variation, the lower is the coefficient of determination. In this case, 25.2% of the total variation in the dependent variable is error or residual variation.

The final observation that relates the correlation coefficient, the coefficient of determination, and the standard error of estimate is to show the relationship between the standard error of estimate and SSE. By substituting [SSE Residual or Error Sum of Squares = SSE = $\Sigma(y - \hat{y})^2$] into the formula for the standard error of estimate, we find:

STANDARD ERROR OF ESTIMATE

$$s_{y \cdot x} = \sqrt{\frac{\text{SSE}}{n - 2}} \qquad [13\text{–}9]$$

Note that $s_{y \cdot x}$ can also be computed using the residual mean square from the ANOVA table.

STANDARD ERROR OF THE ESTIMATE

$$s_{y \cdot x} = \sqrt{\textit{Residual mean square}} \qquad [13\text{–}10]$$

In sum, regression analysis provides two statistics to evaluate the predictive ability of a regression equation: the standard error of the estimate and the coefficient of determination. When reporting the results of a regression analysis, the findings must be clearly explained, especially when using the results to make predictions of the dependent variable. The report must always include a statement regarding the coefficient of determination so that the relative precision of the prediction is known to the reader of the report. Objective reporting of statistical analysis is required so that the readers can make their own decisions.

EXERCISES

29. Given the following ANOVA table:

Source	*DF*	SS	MS	*F*
Regression	1	1000.0	1000.0	26.00
Error	13	500.0	38.46	
Total	14	1500.0		

a. Determine the coefficient of determination.
b. Assuming a direct relationship between the variables, what is the correlation coefficient?
c. Determine the standard error of estimate.

30. On the first statistics exam, the coefficient of determination between the hours studied and the grade earned was 80%. The standard error of estimate was 10. There were 20 students in the class. Develop an ANOVA table for the regression analysis of hours studied as a predictor of the grade earned on the first statistics exam.

LO13-6
Calculate and interpret confidence and prediction intervals.

INTERVAL ESTIMATES OF PREDICTION

The standard error of estimate and the coefficient of determination are two statistics that provide an overall evaluation of the ability of a regression equation to predict a dependent variable. Another way to report the ability of a regression equation to predict is specific to a stated value of the independent variable. For example, we can predict the number of copiers sold (y) for a selected value of number of sales calls made (x). In fact, we can calculate a confidence interval for the predicted value of the dependent variable for a selected value of the independent variable.

STATISTICS IN ACTION

Studies indicate that for both men and women, those who are considered good looking earn higher wages than those who are not. In addition, for men there is a correlation between height and salary. For each additional inch of height, a man can expect to earn an additional $250 per year. So a man 6′6″ tall receives a $3,000 "stature" bonus over his 5′6″ counterpart. Being overweight or underweight is also related to earnings, particularly among women. A study of young women showed the heaviest 10% earned about 6% less than their lighter counterparts.

Assumptions Underlying Linear Regression

Before we present the confidence intervals, the assumptions for properly applying linear regression should be reviewed. Chart 13–13 illustrates these assumptions.

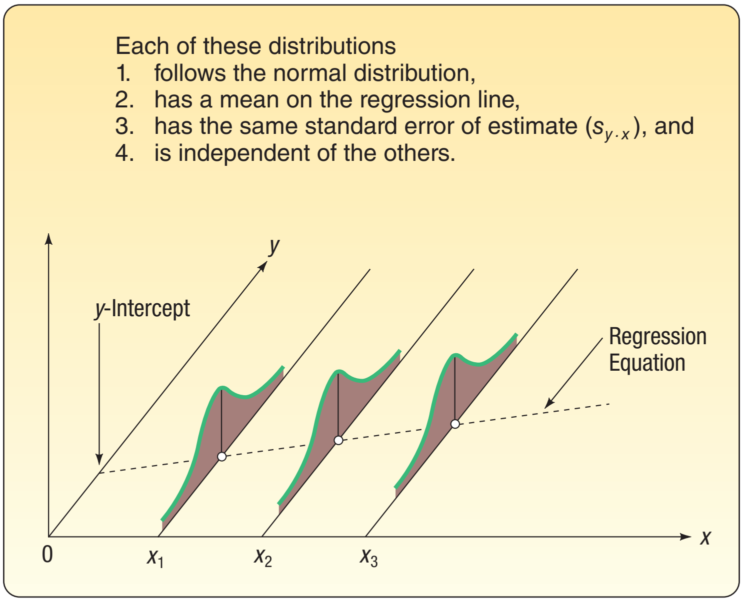


CHART 13–13 Regression Assumptions Shown Graphically

1. For each value of x, there are corresponding y values. These y values follow the normal distribution.
2. The means of these normal distributions lie on the regression line.
3. The standard deviations of these normal distributions are all the same. The best estimate we have of this common standard deviation is the standard error of estimate ($s_{y \cdot x}$).
4. The y values are statistically independent. This means that in selecting a sample, a particular x does not depend on any other value of x. This assumption is particularly important when data are collected over a period of time. In such situations, the errors for a particular time period are often correlated with those of other time periods.

Recall from Chapter 7 that if the values follow a normal distribution, then the mean plus or minus one standard deviation will encompass 68% of the observations, the mean plus or minus two standard deviations will encompass 95% of the observations, and the mean plus or minus three standard deviations will encompass virtually all of the observations. The same relationship exists between the predicted values $\hat{y}$ and the standard error of estimate ($s_{y \cdot x}$).

1. $\hat{y} \pm s_{y \cdot x}$ will include the middle 68% of the observations.
2. $\hat{y} \pm 2s_{y \cdot x}$ will include the middle 95% of the observations.
3. $\hat{y} \pm 3s_{y \cdot x}$ will include virtually all the observations.

We can now relate these assumptions to North American Copier Sales, where we studied the relationship between the number of sales calls and the number of copiers sold. If we drew a parallel line 6.72 units above the regression line and another 6.72 units below the regression line, about 68% of the points would fall between the two lines. Similarly, a line 13.44 [$2s_{y \cdot x} = 2(6.72)$] units above the regression line and another 13.44 units below the regression line should include about 95% of the data values.

As a rough check, refer to column E in the Excel spreadsheet appearing on page 456. Four of the 15 deviations exceed one standard error of estimate. That is, the deviations of Carlos Ramirez, Mark Reynolds, Mike Keil, and Tom Keller all exceed 6.72 (one standard error). All values are less than 13.44 units away from the regression line. In short, 11 of the 15 deviations are within one standard error and all are within two standard errors. That is a fairly good result for a relatively small sample.

Constructing Confidence and Prediction Intervals

When using a regression equation, two different predictions can be made for a selected value of the independent variable. The differences are subtle but very important and are related to the assumptions stated in the last section. Recall that for any selected value of the independent variable (X), the dependent variable (Y) is a random variable that is normally distributed with a mean $\hat{Y}$. Each distribution of Y has a standard deviation equal to the regression analysis's standard error of estimate.

The first interval estimate is called a **confidence interval.** This is used when the regression equation is used to predict the mean value of Y for a given value of x. For example, we would use a confidence interval to estimate the mean salary of all executives in the retail industry based on their years of experience. To determine the confidence interval for the mean value of y for a given x, the formula is:

CONFIDENCE INTERVAL FOR THE MEAN OF Y, GIVEN X

$$\hat{y} \pm t s_{y \cdot x} \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{\Sigma(x - \bar{x})^2}} \qquad \textbf{[13–11]}$$

The second interval estimate is called a prediction interval. This is used when the regression equation is used to predict an individual y for a given value of x. For example, we would estimate the salary of a particular retail executive who has 20 years of experience. To calculate a prediction interval, formula 13-11 is modified by adding a 1 under the radical. To determine the prediction interval for an estimate of an individual for a given x, the formula is:

PREDICTION INTERVAL FOR Y, GIVEN X

$$\hat{y} \pm t s_{y \cdot x} \sqrt{1 + \frac{1}{n} + \frac{(x - \bar{x})^2}{\Sigma(x - \bar{x})^2}} \qquad \textbf{[13–12]}$$

EXAMPLE

We return to the North American Copier Sales illustration. Determine a 95% confidence interval for all sales representatives who make 50 calls, and determine a prediction interval for Sheila Baker, a West Coast sales representative who made 50 calls.

SOLUTION

We use formula (13–11) to determine a confidence level. Table 13–4 includes the necessary totals and a repeat of the information of Table 13–2.

TABLE 13–4 Determining Confidence and Prediction Intervals

Sales Representative	Sales Calls (x)	Copiers Sold (y)	$(x - \bar{x})$	$(x - \bar{x})^2$
Brian Virost	96	41	0	0
Carlos Ramirez	40	41	−56	3,136
Carol Saia	104	51	8	64
Greg Fish	128	60	32	1,024
Jeff Hall	164	61	68	4,624
Mark Reynolds	76	29	−20	400
Meryl Rumsey	72	39	−24	576
Mike Kiel	80	50	−16	256
Ray Snarsky	36	28	−60	3,600
Rich Niles	84	43	−12	144
Ron Broderick	180	70	84	7,056
Sal Spina	132	56	36	1,296
Sani Jones	120	45	24	576
Susan Welch	44	31	−52	2,704
Tom Keller	84	30	−12	144
Total	1440	675	0	25,600

The first step is to determine the number of copiers we expect a sales representative to sell if he or she makes 50 calls. It is 33.0032, found by

$$\hat{y} = 19.9632 + 0.2608x = 19.9632 + 0.2608\,(50) = 33.0032$$

To find the t value, we need to first know the number of degrees of freedom. In this case, the degrees of freedom are $n - 2 = 15 - 2 = 13$. We set the confidence level at 95%. To find the value of t, move down the left-hand column of Appendix B.5 to 13 degrees of freedom, then move across to the column with the 95% level of confidence. The value of t is 2.160.

In the previous section, we calculated the standard error of estimate to be 6.720. We let $x = 50$, and from Table 13–4, the mean number of sales calls is 96.0 (1440/15) and $\Sigma(x - \bar{x})^2 = 25,600$ Inserting these values in formula (13–11), we can determine the confidence interval.

$$\text{Confidence Interval} = \hat{y} \pm t s_{y \cdot x}\sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{\Sigma(x - \bar{x})^2}}$$

$$= 33.0032 \pm 2.160(6.720)\sqrt{\frac{1}{15} + \frac{(50 - 96)^2}{25{,}600}}$$

$$= 33.0032 \pm 5.6090$$

Thus, the 95% confidence interval for all sales representatives who make 50 calls is from 27.3942 up to 38.6122. To interpret, let's round the values. If a sales representative makes 50 calls, he or she can expect to sell 33 copiers. It is likely the sales will range from 27.4 to 38.6 copiers.

Suppose we want to estimate the number of copiers sold by Sheila Baker, who made 50 sales calls. Using formula (13–12), the 95% prediction interval is determined as follows:

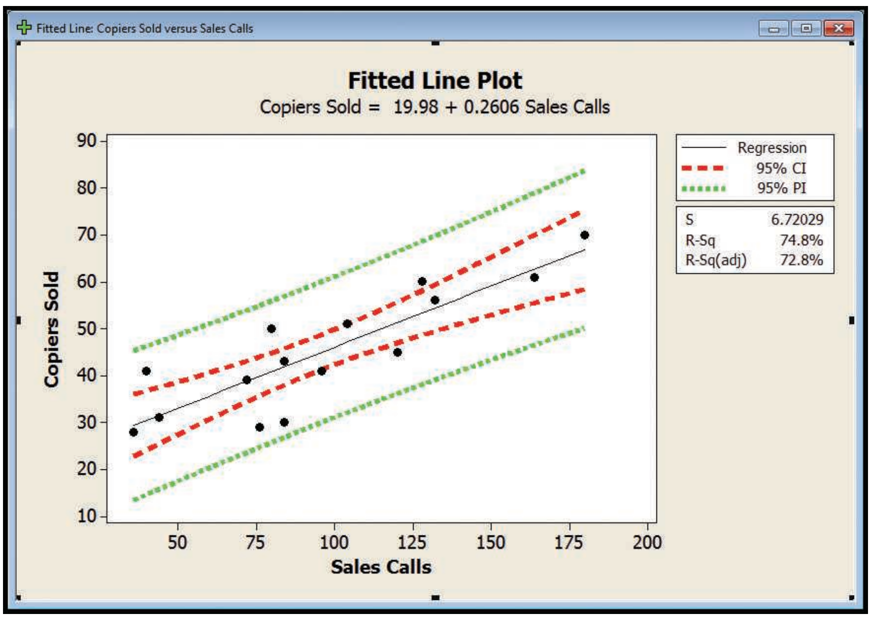
$$\text{Prediction Interval} = \hat{y} \pm ts_{y\cdot x}\sqrt{1 + \frac{1}{n} + \frac{(x - \bar{x})^2}{\Sigma(x - \bar{x})^2}}$$

$$= 33.0032 \pm 2.160(6.720)\sqrt{1 + \frac{1}{15} + \frac{(50 - 96)^2}{25{,}600}}$$

$$= 33.0032 \pm 15.5612$$

Thus, the interval is from 17.442 up to 48.5644 copiers. We conclude that the number of office machines sold will be between about 17.4 and 48.6 for a particular sales representative, such as Sheila Baker, who makes 50 calls. This interval is quite large. It is much larger than the confidence interval for all sales representatives who made 50 calls. It is logical, however, that there should be more variation in the sales estimate for an individual than for a group.

The following Minitab graph shows the relationship between the least squares regression line (in the center), the confidence interval (shown in crimson), and the prediction interval (shown in green). The bands for the prediction interval are always further from the regression line than those for the confidence interval. Also, as the values of x move away from the mean number of calls (96) in either direction, the confidence interval and prediction interval bands widen. This is caused by the numerator of the right-hand term under the radical in formulas (13–11) and (13–12). That is, as the term increases, the widths of the confidence interval and the prediction interval also increase. To put it another way, there is less precision in our estimates as we move away, in either direction, from the mean of the independent variable.



We wish to emphasize again the distinction between a confidence interval and a prediction interval. A confidence interval refers to the mean of all cases for a given value of x and is computed by formula (13–11). A prediction interval refers to a particular, single case for a given value of x and is computed using formula (13–12). The prediction interval will always be wider because of the extra 1 under the radical in the second equation.

SELF-REVIEW 13–6

Refer to the sample data in Self-Review 13–1, where the owner of Haverty's Furniture was studying the relationship between sales and the amount spent on advertising. The advertising expense and sales revenue, both in millions of dollars, for the last 4 months are repeated below.

Month	Advertising Expense ($ million)	Sales Revenue ($ million)
July	2	7
August	1	3
September	3	8
October	4	10

The regression equation was computed to be $\hat{y} = 1.5 + 2.2x$ and the standard error 0.9487. Both variables are reported in millions of dollars. Determine the 90% confidence interval for the typical month in which $3 million was spent on advertising.

EXERCISES

31. Refer to Exercise 13.
- **a.** Determine the .95 confidence interval for the mean predicted when $x = 7$.
- **b.** Determine the .95 prediction interval for an individual predicted when $x = 7$.

32. Refer to Exercise 14.
- **a.** Determine the .95 confidence interval for the mean predicted when $x = 7$.
- **b.** Determine the .95 prediction interval for an individual predicted when $x = 7$.

33. Refer to Exercise 15.
- **a.** Determine the .95 confidence interval, in thousands of kilowatt-hours, for the mean of all six-room homes.
- **b.** Determine the .95 prediction interval, in thousands of kilowatt-hours, for a particular six-room home.

34. Refer to Exercise 16.
- **a.** Determine the .95 confidence interval, in thousands of dollars, for the mean of all sales personnel who make 40 contacts.
- **b.** Determine the .95 prediction interval, in thousands of dollars, for a particular salesperson who makes 40 contacts.

TRANSFORMING DATA

LO13-7
Use a log function to transform a nonlinear relationship.

Regression analysis describes the relationship between two variables. A requirement is that this relationship be linear. The same is true of the correlation coefficient. It measures the strength of a linear relationship between two variables. But what if the relationship is not linear? The remedy is to rescale one or both of the variables so the new relationship is linear. For example, instead of using the actual values of the dependent variable, y, we would create a new dependent variable by computing the log to the base 10 of y, Log(y). This calculation is called a transformation. Other common transformations include taking the square root, taking the reciprocal, or squaring one or both of the variables.

Thus, two variables could be closely related, but their relationship is not linear. Be cautious when you are interpreting the correlation coefficient or a regression equation. These statistics may indicate there is no linear relationship, but there could be a relationship of some other nonlinear or curvilinear form. The following example explains the details.

EXAMPLE

GroceryLand Supermarkets is a regional grocery chain with over 300 stores located in the midwestern United States. The corporate director of marketing for GroceryLand wishes to study the effect of price on the weekly sales of two-liter bottles of their private-brand diet cola. The objectives of the study are:

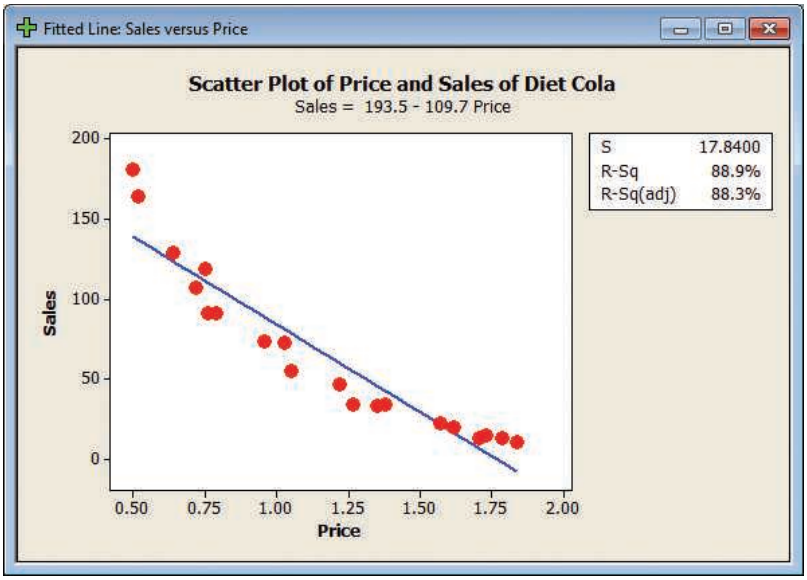
1. To determine whether there is a relationship between selling price and weekly sales. Is this relationship direct or indirect? Is it strong or weak?
2. To determine the effect of price increases or decreases on sales. Can we effectively forecast sales based on the price?

SOLUTION

To begin the project, the marketing director meets with the vice president of sales and other company staff members. They decide that it would be reasonable to price the two-liter bottle of their private-brand diet cola from $0.50 up to $2.00. To collect the data needed to analyze the relationship between price and sales, the marketing director selects a random sample of 20 stores and then randomly assigns a selling price for the two-liter bottle of diet cola between $0.50 and $2.00 to each selected store. The director contacts each of the 20 store managers included in the study to tell them the selling price and ask them to report the sales for the product at the end of the week. The results are reported below. For example, store number A-17 sold 181 two-liter bottles of diet cola at $0.50 each.

GroceryLand Sales and Price Data			GroceryLand Sales and Price Data		
Store Number	Price	Sales	Store Number	Price	Sales
A-17	0.50	181	A-30	0.76	91
A-121	1.35	33	A-127	1.79	13
A-227	0.79	91	A-266	1.57	22
A-135	1.71	13	A-117	1.27	34
A-6	1.38	34	A-132	0.96	74
A-282	1.22	47	A-120	0.52	164
A-172	1.03	73	A-272	0.64	129
A-296	1.84	11	A-120	1.05	55
A-143	1.73	15	A-194	0.72	107
A-66	1.62	20	A-105	0.75	119

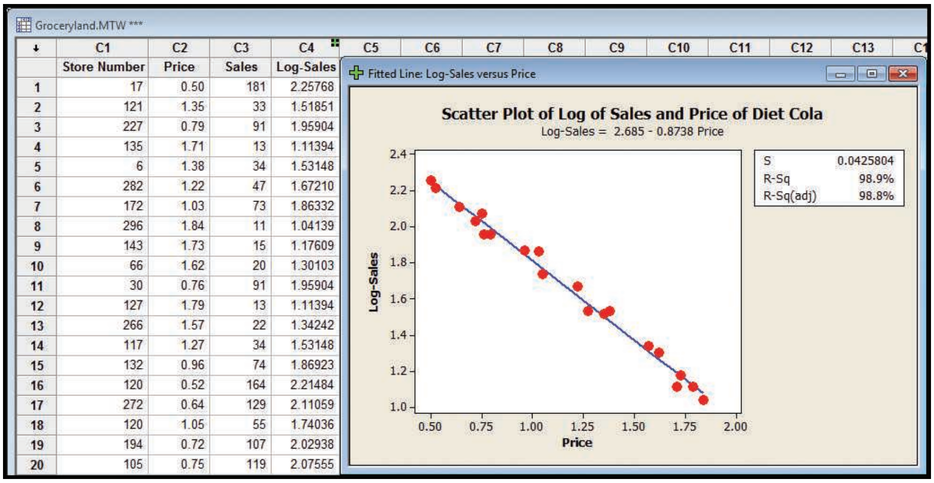
To examine the relationship between Price and Sales, we use regression analysis setting *Price* as the independent variable and *Sales* as the dependent variable. The analysis will provide important information about the relationship between the variables. The analysis is summarized in the following Minitab output.



From the output, we can make these conclusions:

1. The relationship between the two variables is inverse or indirect. As the *Price* of the cola increases, the *Sales* of the product decreases. Given basic economic theory of price and demand, this is expected.
2. There is a strong relationship between the two variables. The coefficient of determination is 88.9%. So 88.9% of the variation in *Sales* is accounted for by the variation in *Price*. From the coefficient of determination, we can compute the correlation coefficient as the square root of the coefficient of determination. The correlation coefficient is the square root of 0.889, or 0.943. The sign of the correlation coefficient is negative because sales are inversely related to price. Therefore, the correlation coefficient is −0.943.
3. Before continuing our summary of conclusions, we should look carefully at the scatter diagram and the plot of the regression line. The assumption of a linear relationship is tenuous. If the relationship is linear, the data points should be distributed both above and below the line over the entire range of the independent variable. However, for the highest and lowest prices, the data points are above the regression line. For the selling prices in the middle, most of the data points are below the regression line. So the linear regression equation does not effectively describe the relationship between *Price* and *Sales.* A transformation of the data is needed to create a linear relationship.

By transforming one of the variables, we may be able to change the nonlinear relationship between the variables to a linear relationship. Of the possible choices, the director of marketing decides to transform the dependent variable, *Sales,* by taking the logarithm to the base 10 of each *Sales* value. Note the new variable, *Log-Sales,* in the following analysis. Now, the regression analysis uses *Log-Sales* as the dependent variable and *Price* as the independent variable. This analysis is reported below.



	C1 Store Number	C2 Price	C3 Sales	C4 Log-Sales
1	17	0.50	181	2.25768
2	121	1.35	33	1.51851
3	227	0.79	91	1.95904
4	135	1.71	13	1.11394
5	6	1.38	34	1.53148
6	282	1.22	47	1.67210
7	172	1.03	73	1.86332
8	296	1.84	11	1.04139
9	143	1.73	15	1.17609
10	66	1.62	20	1.30103
11	30	0.76	91	1.95904
12	127	1.79	13	1.11394
13	266	1.57	22	1.34242
14	117	1.27	34	1.53148
15	132	0.96	74	1.86923
16	120	0.52	164	2.21484
17	272	0.64	129	2.11059
18	120	1.05	55	1.74036
19	194	0.72	107	2.02938
20	105	0.75	119	2.07555

What can we conclude from the regression analysis using the transformation of the dependent variable *Sales?*

1. By transforming the dependent variable, *Sales,* we increase the coefficient of determination from 0.889 to 0.989. So *Price* now explains nearly all of the variation in *Log-Sales.*
2. Compare this result with the scatter diagram before we transformed the dependent variable. The transformed data seem to fit the linear relationship requirement much better. Observe that the data points are both above and below the regression line over the range of *Price.*
3. The regression equation is $\hat{y} = 2.685 - 0.8738x$. The sign of the slope value is negative, confirming the inverse association between the variables. We can use the new equation to estimate sales and study the effect of changes in price. For example, if we decided to sell the two-liter bottle of diet cola for \$1.25, the predicted *Log-Sales* is:

$$\hat{y} = 2.685 - 0.8738x = 2.685 - 0.8738\,(1.25) = 1.593$$

Remember that the regression equation now predicts the log, base10, of *Sales.* Therefore, we must undo the transformation by taking the antilog of 1.593, which is $10^{1.593}$, or 39.174. So, if we price the two-liter diet cola product at \$1.25, the predicted weekly sales are 39 bottles. If we increase the price to \$2.00, the regression equation would predict a value of .9374. Taking the antilog, $10^{.9374}$, the predicted sales decrease to 8.658, or, rounding, 9 two-liter bottles per week. Clearly, as price increases, sales decrease. This relationship will be very helpful to GroceryLand when making pricing decisions for this product.

EXERCISES

35. **FILE** Given the following sample of five observations, develop a scatter diagram, using *x* as the independent variable and *y* as the dependent variable, and compute the correlation coefficient. Does the relationship between the variables appear to be linear? Try squaring the *x* variable and then develop a scatter diagram and determine the correlation coefficient. Summarize your analysis.

x	−8	−16	12	2	18
y	58	247	153	3	341

36. **FILE** Every April, The Masters—one of the most prestigious golf tournaments on the PGA golf tour—is played in Augusta, Georgia. In 2016, 55 players received prize money. The 2016 winner, Danny Willett, earned a prize of $1,800,000. Jordan Spieth and Lee Westwood tied for second place, earning $880,000. Two amateur players finished "in the money" but they could not accept the prize money. They are not included in the data. The data are briefly summarized below. Each player has three corresponding variables: finishing position, score, and prize (in dollars). We want to study the relationship between score and prize.

Position	Player	Score	Prize
1	Danny Willett	283	$1,800,000
2(tied)	Jordan Spieth	286	$880,000
2(tied)	Lee Westwood	286	$880,000
4(tied)	Paul Casey	287	$413,333
4(tied)	J.B. Holmes	287	$413,333
4(tied)	Dustin Johnson	287	$413,333
..	..	..	..
..	..	..	..
..	..	..	..
52(tied)	Keegan Bradley	301	$24,900
52(tied)	Larry Mize	301	$24,900
54	Hunter Mahan	302	$24,000
55(tied)	Kevin Na	303	$23,400
55(tied)	Cameron Smith	303	$23,400
57	Thongchai Jaidee	307	$23,000

a. Using *Score* as the independent variable and *Prize* as the dependent variable, develop a scatter diagram. Does the relationship appear to be linear? Does it seem reasonable that as *Score* increases the *Prize* decreases?

b. What percentage of the variation in the dependent variable, *Prize,* is accounted for by the independent variable, *Score*?

c. Calculate a new variable, *Log-Prize*, computing the log to the base 10 of *Prize*. Draw a scatter diagram with *Log-Prize* as the dependent variable and *Score* as the independent variable.

d. Develop a regression equation and compute the coefficient of determination using *Log-Prize* as the dependent variable.

e. Compare the coefficient of determination in parts (b) and (d). What do you conclude?

f. Write out the regression equation developed in part (d). If a player shot an even par score of 288 for the four rounds, how much would you expect that player to earn?

CHAPTER SUMMARY

I. A scatter diagram is a graphic tool used to portray the relationship between two variables.
- **A.** The dependent variable is scaled on the *Y*-axis and is the variable being estimated.
- **B.** The independent variable is scaled on the *X*-axis and is the variable used as the predictor.

II. The correlation coefficient measures the strength of the linear association between two variables.
- **A.** Both variables must be at least the interval scale of measurement.
- **B.** The correlation coefficient can range from -1.00 to 1.00.

C. If the correlation between the two variables is 0, there is no association between them.
D. A value of 1.00 indicates perfect positive correlation, and a value of −1.00 indicates perfect negative correlation.
E. A positive sign means there is a direct relationship between the variables, and a negative sign means there is an inverse relationship.
F. It is designated by the letter r and found by the following equation:

$$r = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{(n - 1)s_x s_y} \quad \textbf{[13–1]}$$

G. To test a hypothesis that a population correlation is different from 0, we use the following statistic:

$$t = \frac{r\sqrt{n - 2}}{\sqrt{1 - r^2}} \quad \text{with } n - 2 \text{ degrees of freedom} \quad \textbf{[13–2]}$$

III. In regression analysis, we estimate one variable based on another variable.
A. The variable being estimated is the dependent variable.
B. The variable used to make the estimate or predict the value is the independent variable.
1. The relationship between the variables is linear.
2. Both the independent and the dependent variables must be interval or ratio scale.
3. The least squares criterion is used to determine the regression equation.
IV. The least squares regression line is of the form $\hat{y} = a + bx$
A. $\hat{y}$ is the estimated value of y for a selected value of x.
B. a is the constant or intercept.
1. It is the value of $\hat{y}$ when $x = 0$.
2. a is computed using the following equation.

$$a = \bar{y} - b\bar{x} \quad \textbf{[13–5]}$$

C. b is the slope of the fitted line.
1. It shows the amount of change in $\hat{y}$ for a change of one unit in x.
2. A positive value for b indicates a direct relationship between the two variables. A negative value indicates an inverse relationship.
3. The sign of b and the sign of r, the correlation coefficient, are always the same.
4. b is computed using the following equation.

$$b = r\left(\frac{s_y}{s_x}\right) \quad \textbf{[13–4]}$$

D. x is the value of the independent variable.
V. For a regression equation, the slope is tested for significance.
A. We test the hypothesis that the slope of the line in the population is 0.
1. If we do not reject the null hypothesis, we conclude there is no relationship between the two variables.
2. The test is equivalent to the test for the correlation coefficient.
B. When testing the null hypothesis about the slope, the test statistic is with $n - 2$ degrees of freedom:

$$t = \frac{b - 0}{s_b} \quad \textbf{[13–6]}$$

VI. The standard error of estimate measures the variation around the regression line.
A. It is in the same units as the dependent variable.
B. It is based on squared deviations from the regression line.
C. Small values indicate that the points cluster closely about the regression line.
D. It is computed using the following formula.

$$s_{y \cdot x} = \sqrt{\frac{\Sigma(y - \hat{y})^2}{n - 2}} \quad \textbf{[13–7]}$$

VII. The coefficient of determination is the proportion of the variation of a dependent variable explained by the independent variable.
  A. It ranges from 0 to 1.0.
  B. It is the square of the correlation coefficient.
  C. It is found from the following formula.

$$r^2 = \frac{\text{SSR}}{\text{SS Total}} = 1 - \frac{\text{SSE}}{\text{SS Total}} \qquad \textbf{[13–8]}$$

VIII. Inference about linear regression is based on the following assumptions.
  A. For a given value of x, the values of Y are normally distributed about the line of regression.
  B. The standard deviation of each of the normal distributions is the same for all values of x and is estimated by the standard error of estimate.
  C. The deviations from the regression line are independent, with no pattern to the size or direction.

IX. There are two types of interval estimates.
  A. In a confidence interval, the mean value of y is estimated for a given value of x.
    1. It is computed from the following formula.

$$\hat{y} \pm t s_{y \cdot x} \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{\Sigma(x - \bar{x})^2}} \qquad \textbf{[13–11]}$$

    2. The width of the interval is affected by the level of confidence, the size of the standard error of estimate, and the size of the sample, as well as the value of the independent variable.
  B. In a prediction interval, the individual value of y is estimated for a given value of x.
    1. It is computed from the following formula.

$$\hat{y} \pm t s_{y \cdot x} \sqrt{1 + \frac{1}{n} + \frac{(x - \bar{x})^2}{\Sigma(x - \bar{x})^2}} \qquad \textbf{[13–12]}$$

    2. The difference between formulas (13–11) and (13–12) is the 1 under the radical.
      a. The prediction interval will be wider than the confidence interval.
      b. The prediction interval is also based on the level of confidence, the size of the standard error of estimate, the size of the sample, and the value of the independent variable.

PRONUNCIATION KEY

SYMBOL	MEANING	PRONUNCIATION
Σxy	Sum of the products of x and y	*Sum x y*
ρ	Correlation coefficient in the population	*Rho*
$\hat{y}$	Estimated value of Y	*y hat*
$s_{y \cdot x}$	Standard error of estimate	*s sub y dot x*
r^2	Coefficient of determination	*r square*

CHAPTER EXERCISES

37. A regional commuter airline selected a random sample of 25 flights and found that the correlation between the number of passengers and the total weight, in pounds, of luggage stored in the luggage compartment is 0.94. Using the .05 significance level, can we conclude that there is a positive association between the two variables?

38. A sociologist claims that the success of students in college (measured by their GPA) is related to their family's income. For a sample of 20 students, the correlation coefficient is 0.40. Using the 0.01 significance level, can we conclude that there is a positive correlation between the variables?

39. An Environmental Protection Agency study of 12 automobiles revealed a correlation of 0.47 between engine size and emissions. At the .01 significance level, can we conclude that there is a positive association between these variables? What is the *p*-value? Interpret.

40. **FILE** A suburban hotel derives its revenue from its hotel and restaurant operations. The owners are interested in the relationship between the number of rooms occupied on a nightly basis and the revenue per day in the restaurant. Below is a sample of 25 days (Monday through Thursday) from last year showing the restaurant income and number of rooms occupied.

Day	Revenue	Occupied	Day	Revenue	Occupied
1	$1,452	23	14	$1,425	27
2	1,361	47	15	1,445	34
3	1,426	21	16	1,439	15
4	1,470	39	17	1,348	19
5	1,456	37	18	1,450	38
6	1,430	29	19	1,431	44
7	1,354	23	20	1,446	47
8	1,442	44	21	1,485	43
9	1,394	45	22	1,405	38
10	1,459	16	23	1,461	51
11	1,399	30	24	1,490	61
12	1,458	42	25	1,426	39
13	1,537	54			

Use a statistical software package to answer the following questions.

a. Does the revenue seem to increase as the number of occupied rooms increases? Draw a scatter diagram to support your conclusion.

b. Determine the correlation coefficient between the two variables. Interpret the value.

c. Is it reasonable to conclude that there is a positive relationship between revenue and occupied rooms? Use the .10 significance level.

d. What percent of the variation in revenue in the restaurant is accounted for by the number of rooms occupied?

41. **FILE** The table below shows the number of cars (in millions) sold in the United States for various years and the percent of those cars manufactured by GM.

Year	Cars Sold (millions)	Percent GM	Year	Cars Sold (millions)	Percent GM
1950	6.0	50.2	1985	15.4	40.1
1955	7.8	50.4	1990	13.5	36.0
1960	7.3	44.0	1995	15.5	31.7
1965	10.3	49.9	2000	17.4	28.6
1970	10.1	39.5	2005	16.9	26.9
1975	10.8	43.1	2010	11.6	19.1
1980	11.5	44.0	2015	17.5	17.6

Use a statistical software package to answer the following questions.

a. Is the number of cars sold directly or indirectly related to GM's percentage of the market? Draw a scatter diagram to show your conclusion.

b. Determine the correlation coefficient between the two variables. Interpret the value.

c. Is it reasonable to conclude that there is a negative association between the two variables? Use the .01 significance level.

d. How much of the variation in GM's market share is accounted for by the variation in cars sold?

42. For a sample of 40 large U.S. cities, the correlation between the mean number of square feet per office worker and the mean monthly rental rate in the central business district is −0.363. At the .05 significance level, can we conclude that there is a negative association between the two variables.

43. **FILE** For each of the 32 National Football League teams, the numbers of points scored and allowed during the 2016 season are shown below.

TEAM	Conference	PTS Scored	PTS Allowed	TEAM	Conference	PTS Scored	PTS Allowed
Baltimore	AFC	343	321	Arizona	NFC	418	362
Buffalo	AFC	399	378	Atlanta	NFC	540	406
Cincinnati	AFC	325	315	Carolina	NFC	369	402
Cleveland	AFC	264	452	Chicago	NFC	279	399
Denver	AFC	333	297	Dallas	NFC	421	306
Houston	AFC	279	328	Detroit	NFC	346	358
Indianapolis	AFC	411	392	Green Bay	NFC	432	388
Jacksonville	AFC	318	400	Los Angeles	NFC	224	394
Kansas City	AFC	389	311	Minnesota	NFC	327	307
Miami	AFC	363	380	NY Giants	NFC	469	454
New England	AFC	441	250	New Orleans	NFC	310	284
NY Jets	AFC	275	409	Philadelphia	NFC	367	331
Oakland	AFC	416	385	San Francisco	NFC	309	480
Pittsburgh	AFC	399	327	Seattle	NFC	354	292
San Diego	AFC	410	423	Tampa Bay	NFC	354	369
Tennessee	AFC	381	378	Washington	NFC	396	383

Assuming these are sample data, answer the following questions. You may use statistical software to assist you.

a. What is the correlation coefficient between these variables? Are you surprised the association is negative? Interpret your results.

b. Find the coefficient of determination. What does it say about the relationship?

c. At the .05 significance level, can you conclude there is a negative association between "points scored" and "points allowed"?

d. At the .05 significance level, can you conclude there is a negative association between "points scored" and "points allowed" for each conference?

44. **FILE** The Cotton Mill is an upscale chain of women's clothing stores, located primarily in the southwest United States. Due to recent success, The Cotton Mill's top management is planning to expand by locating new stores in other regions of the country. The director of planning has been asked to study the relationship between yearly sales and the store size. As part of the study, the director selects a sample of 25 stores and determines the size of the store in square feet and the sales for last year. The sample data follow. The use of statistical software is suggested.

Store Size (thousands of square feet)	Sales (millions $)	Store Size (thousands of square feet)	Sales (millions $)
3.7	9.18	0.4	0.55
2.0	4.58	4.2	7.56
5.0	8.22	3.1	2.23
0.7	1.45	2.6	4.49
2.6	6.51	5.2	9.90
2.9	2.82	3.3	8.93
5.2	10.45	3.2	7.60
5.9	9.94	4.9	3.71
3.0	4.43	5.5	5.47
2.4	4.75	2.9	8.22
2.4	7.30	2.2	7.17
0.5	3.33	2.3	4.35
5.0	6.76		

a. Draw a scatter diagram. Use store size as the independent variable. Does there appear to be a relationship between the two variables. Is it positive or negative?
b. Determine the correlation coefficient and the coefficient of determination. Is the relationship strong or weak? Why?
c. At the .05 significance level, can we conclude there is a significant positive correlation?

45. FILE The manufacturer of Cardio Glide exercise equipment wants to study the relationship between the number of months since the glide was purchased and the time, in hours, the equipment was used last week.

Person	Months Owned	Hours Exercised	Person	Months Owned	Hours Exercised
Rupple	12	4	Massa	2	8
Hall	2	10	Sass	8	3
Bennett	6	8	Karl	4	8
Longnecker	9	5	Malrooney	10	2
Phillips	7	5	Veights	5	5

a. Plot the information on a scatter diagram. Let hours of exercise be the dependent variable. Comment on the graph.
b. Determine the correlation coefficient. Interpret.
c. At the .01 significance level, can we conclude that there is a negative association between the variables?

46. The following regression equation was computed from a sample of 20 observations:

$$\hat{y} = 15 - 5x$$

SSE was found to be 100 and SS total, 400.
a. Determine the standard error of estimate.
b. Determine the coefficient of determination.
c. Determine the correlation coefficient. (Caution: Watch the sign!)

47. FILE City planners believe that larger cities are populated by older residents. To investigate the relationship, data on population and median age in 10 large cities were collected.

City	Population City (in millions)	Median Age
Chicago, IL	2.833	31.5
Dallas, TX	1.233	30.5
Houston, TX	2.144	30.9
Los Angeles, CA	3.849	31.6
New York, NY	8.214	34.2
Philadelphia, PA	1.448	34.2
Phoenix, AZ	1.513	30.7
San Antonio, TX	1.297	31.7
San Diego, CA	1.257	32.5
San Jose, CA	0.930	32.6

a. Plot these data on a scatter diagram with median age as the dependent variable.
b. Find the correlation coefficient.
c. A regression analysis was performed and the resulting regression equation is Median Age = 31.4 + 0.272 Population. Interpret the meaning of the slope.
d. Estimate the median age for a city of 2.5 million people.
e. Here is a portion of the regression software output. What does it tell you?

```
Predictor        Coef    SE Coef       T        P
Constant      31.3672     0.6158   50.94    0.000
Population     0.2722     0.1901    1.43    0.190
```

f. Using the .10 significance level, test the significance of the slope. Interpret the result. Is there a significant relationship between the two variables?

48. **FILE** Emily Smith decides to buy a fuel-efficient used car. Here are several vehicles she is considering, with the estimated cost to purchase and the age of the vehicle.

Vehicle	Estimated Cost	Age
Honda Insight	$5,555	8
Toyota Prius	$17,888	3
Toyota Prius	$9,963	6
Toyota Echo	$6,793	5
Honda Civic Hybrid	$10,774	5
Honda Civic Hybrid	$16,310	2
Chevrolet Cruz	$2,475	8
Mazda3	$2,808	10
Toyota Corolla	$7,073	9
Acura Integra	$8,978	8
Scion xB	$11,213	2
Scion xA	$9,463	3
Mazda3	$15,055	2
Mini Cooper	$20,705	2

a. Plot these data on a scatter diagram with estimated cost as the dependent variable.
b. Find the correlation coefficient.
c. A regression analysis was performed and the resulting regression equation is Estimated Cost = 18358 − 1534 Age. Interpret the meaning of the slope.
d. Estimate the cost of a five-year-old car.
e. Here is a portion of the regression software output. What does it tell you?

```
Predictor         Coef    SE Coef        T        P
Constant         18358       1817    10.10    0.000
Age            -1533.6      306.3    -5.01    0.000
```

f. Using the .10 significance level, test the significance of the slope. Interpret the result. Is there a significant relationship between the two variables?

49. **FILE** The National Highway Association is studying the relationship between the number of bidders on a highway project and the winning (lowest) bid for the project. Of particular interest is whether the number of bidders increases or decreases the amount of the winning bid.

Project	Number of Bidders, x	Winning Bid ($ millions), y	Project	Number of Bidders, x	Winning Bid ($ millions), y
1	9	5.1	9	6	10.3
2	9	8.0	10	6	8.0
3	3	9.7	11	4	8.8
4	10	7.8	12	7	9.4
5	5	7.7	13	7	8.6
6	10	5.5	14	7	8.1
7	7	8.3	15	6	7.8
8	11	5.5			

a. Determine the regression equation. Interpret the equation. Do more bidders tend to increase or decrease the amount of the winning bid?
b. Estimate the amount of the winning bid if there were seven bidders.
c. A new entrance is to be constructed on the Ohio Turnpike. There are seven bidders on the project. Develop a 95% prediction interval for the winning bid.
d. Determine the coefficient of determination. Interpret its value.

50. **FILE** Mr. William Profit is studying companies going public for the first time. He is particularly interested in the relationship between the size of the offering and the price per share. A sample of 15 companies that recently went public revealed the following information.

Company	Size ($ millions), x	Price per Share, y	Company	Size ($ millions), x	Price per Share, y
1	9.0	10.8	9	160.7	11.3
2	94.4	11.3	10	96.5	10.6
3	27.3	11.2	11	83.0	10.5
4	179.2	11.1	12	23.5	10.3
5	71.9	11.1	13	58.7	10.7
6	97.9	11.2	14	93.8	11.0
7	93.5	11.0	15	34.4	10.8
8	70.0	10.7			

a. Determine the regression equation.
b. Conduct a test to determine whether the slope of the regression line is positive.
c. Determine the coefficient of determination. Do you think Mr. Profit should be satisfied with using the size of the offering as the independent variable?

51. **FILE** Bardi Trucking Co., located in Cleveland, Ohio, makes deliveries in the Great Lakes region, the Southeast, and the Northeast. Jim Bardi, the president, is studying the relationship between the distance a shipment must travel and the length of time, in days, it takes the shipment to arrive at its destination. To investigate, Mr. Bardi selected a random sample of 20 shipments made last month. Shipping distance is the independent variable and shipping time is the dependent variable. The results are as follows:

Shipment	Distance (miles)	Shipping Time (days)	Shipment	Distance (miles)	Shipping Time (days)
1	656	5	11	862	7
2	853	14	12	679	5
3	646	6	13	835	13
4	783	11	14	607	3
5	610	8	15	665	8
6	841	10	16	647	7
7	785	9	17	685	10
8	639	9	18	720	8
9	762	10	19	652	6
10	762	9	20	828	10

a. Draw a scatter diagram. Based on these data, does it appear that there is a relationship between how many miles a shipment has to go and the time it takes to arrive at its destination?
b. Determine the correlation coefficient. Can we conclude that there is a positive correlation between distance and time? Use the .05 significance level.
c. Determine and interpret the coefficient of determination.
d. Determine the standard error of estimate.
e. Would you recommend using the regression equation to predict shipping time? Why or why not?

52. **FILE** Super Markets Inc. is considering expanding into the Scottsdale, Arizona, area. You, as director of planning, must present an analysis of the proposed expansion to the operating committee of the board of directors. As a part of your proposal, you need to include information on the amount people in the region spend per month for grocery items. You would also like to include information on the relationship between the amount spent for grocery items and income. Your assistant gathered the following sample information.

Household	Amount Spent	Monthly Income
1	$ 555	$4,388
2	489	4,558
⋮	⋮	⋮
39	1,206	9,862
40	1,145	9,883

a. Let the amount spent be the dependent variable and monthly income the independent variable. Create a scatter diagram, using a software package.
b. Determine the regression equation. Interpret the slope value.
c. Determine the correlation coefficient. Can you conclude that it is greater than 0?

53. **FILE** Below is information on the price per share and the dividend for a sample of 30 companies.

Company	Price per Share	Dividend
1	$20.00	$ 3.14
2	22.01	3.36
⋮	⋮	⋮
29	77.91	17.65
30	80.00	17.36

a. Calculate the regression equation using selling price based on the annual dividend.
b. Test the significance of the slope.
c. Determine the coefficient of determination. Interpret its value.
d. Determine the correlation coefficient. Can you conclude that it is greater than 0 using the .05 significance level?

54. A highway employee performed a regression analysis of the relationship between the number of construction work-zone fatalities and the number of unemployed people in a state. The regression equation is Fatalities = 12.7 + 0.000114 (Unemp). Some additional output is:

```
Predictor          Coef        SE Coef       T        P
Constant         12.726          8.115    1.57    0.134
Unemp        0.00011386     0.00002896    3.93    0.001
Analysis of Variance
Source             DF       SS       MS       F       P
Regression          1    10354    10354   15.46   0.001
Residual Error     18    12054      670
Total              19    22408
```

a. How many states were in the sample?
b. Determine the standard error of estimate.
c. Determine the coefficient of determination.
d. Determine the correlation coefficient.
e. At the .05 significance level, does the evidence suggest there is a positive association between fatalities and the number unemployed?

55. A regression analysis relating the current market value in dollars to the size in square feet of homes in Greene County, Tennessee, follows. The regression equation is: Value = −37,186 + 65.0 Size.

```
Predictor         Coef      SE Coef       T        P
Constant        −37186         4629   −8.03    0.000
Size            64.993        3.047   21.33    0.000
Analysis of Variance
Source            DF           SS            MS        F        P
Regression         1  13548662082   13548662082   454.98    0.000
Residual Error    33    982687392      29778406
Total             34  14531349474
```

a. How many homes were in the sample?
b. Compute the standard error of estimate.
c. Compute the coefficient of determination.
d. Compute the correlation coefficient.
e. At the .05 significance level, does the evidence suggest a positive association between the market value of homes and the size of the home in square feet?

56. **FILE** The following table shows the mean annual percent return on capital (profitability) and the mean annual percentage sales growth for eight aerospace and defense companies.

Company	Profitability	Growth
Alliant Techsystems	23.1	8.0
Boeing	13.2	15.6
General Dynamics	24.2	31.2
Honeywell	11.1	2.5
L-3 Communications	10.1	35.4
Northrop Grumman	10.8	6.0
Rockwell Collins	27.3	8.7
United Technologies	20.1	3.2

a. Compute the correlation coefficient. Conduct a test of hypothesis to determine if it is reasonable to conclude that the population correlation is greater than zero. Use the .05 significance level.
b. Develop the regression equation for profitability based on growth. Can we conclude that the slope of the regression line is negative?
c. Use a software package to determine the residual for each observation. Which company has the largest residual?

57. **FILE** The following data show the retail price for 12 randomly selected laptop computers along with their corresponding processor speeds in gigahertz.

Computer	Speed	Price	Computer	Speed	Price
1	2.0	1008.50	7	2.0	1098.50
2	1.6	461.00	8	1.6	693.50
3	1.6	532.00	9	2.0	1057.00
4	1.8	971.00	10	1.6	1001.00
5	2.0	1068.50	11	1.0	468.50
6	1.2	506.00	12	1.4	434.50

a. Develop a linear equation that can be used to describe how the price depends on the processor speed.
b. Based on your regression equation, is there one machine that seems particularly over- or underpriced?
c. Compute the correlation coefficient between the two variables. At the .05 significance level, conduct a test of hypothesis to determine if the population correlation is greater than zero.

58. **FILE** A consumer buying cooperative tested the effective heating area of 20 different electric space heaters with different wattages. Here are the results.

Heater	Wattage	Area	Heater	Wattage	Area
1	1,500	205	11	1,250	116
2	750	70	12	500	72
3	1,500	199	13	500	82
4	1,250	151	14	1,500	206
5	1,250	181	15	2,000	245
6	1,250	217	16	1,500	219
7	1,000	94	17	750	63
8	2,000	298	18	1,500	200
9	1,000	135	19	1,250	151
10	1,500	211	20	500	44

a. Compute the correlation between the wattage and heating area. Is there a direct or an indirect relationship?
b. Conduct a test of hypothesis to determine if it is reasonable that the coefficient is greater than zero. Use the .05 significance level.
c. Develop the regression equation for effective heating based on wattage.
d. Which heater looks like the "best buy" based on the size of the residual?

59. FILE A dog trainer is exploring the relationship between the size of the dog (weight in pounds) and its daily food consumption (measured in standard cups). Below is the result of a sample of 18 observations.

Dog	Weight	Consumption	Dog	Weight	Consumption
1	41	3	10	91	5
2	148	8	11	109	6
3	79	5	12	207	10
4	41	4	13	49	3
5	85	5	14	113	6
6	111	6	15	84	5
7	37	3	16	95	5
8	111	6	17	57	4
9	41	3	18	168	9

a. Compute the correlation coefficient. Is it reasonable to conclude that the correlation in the population is greater than zero? Use the .05 significance level.
b. Develop the regression equation for cups based on the dog's weight. How much does each additional cup change the estimated weight of the dog?
c. Is one of the dogs a big undereater or overeater?

60. Waterbury Insurance Company wants to study the relationship between the amount of fire damage and the distance between the burning house and the nearest fire station. This information will be used in setting rates for insurance coverage. For a sample of 30 claims for the last year, the director of the actuarial department determined the distance from the fire station (x) and the amount of fire damage, in thousands of dollars (y). The MegaStat output is reported below.

```
ANOVA table
Source                SS    df          MS       F
Regression    1,864.5782     1  1,864.5782   38.83
Residual      1,344.4934    28     48.0176
Total         3,209.0716    29
Regression output
Variables     Coefficients    Std. Error  t(df = 28)
Intercept          12.3601        3.2915       3.755
Distance–X          4.7956        0.7696       6.231
```

Answer the following questions.
a. Write out the regression equation. Is there a direct or indirect relationship between the distance from the fire station and the amount of fire damage?
b. How much damage would you estimate for a fire 5 miles from the nearest fire station?
c. Determine and interpret the coefficient of determination.
d. Determine the correlation coefficient. Interpret its value. How did you determine the sign of the correlation coefficient?
e. Conduct a test of hypothesis to determine if there is a significant relationship between the distance from the fire station and the amount of damage. Use the .01 significance level and a two-tailed test.

61. **FILE** TravelAir.com samples domestic airline flights to explore the relationship between airfare and distance. The service would like to know if there is a correlation between airfare and flight distance. If there is a correlation, what percentage of the variation in airfare is accounted for by distance? How much does each additional mile add to the fare? The data follow.

Origin	Destination	Distance	Fare
Detroit, MI	Myrtle Beach, SC	636	$109
Baltimore, MD	Sacramento, CA	2,395	252
Las Vegas, NV	Philadelphia, PA	2,176	221
Sacramento, CA	Seattle, WA	605	151
Atlanta, GA	Orlando, FL	403	138
Boston, MA	Miami, FL	1,258	209
Chicago, IL	Covington, KY	264	254
Columbus, OH	Minneapolis, MN	627	259
Fort Lauderdale, FL	Los Angeles, CA	2,342	215
Chicago, IL	Indianapolis, IN	177	128
Philadelphia, PA	San Francisco, CA	2,521	348
Houston, TX	Raleigh/Durham, NC	1,050	224
Houston, TX	Midland/Odessa, TX	441	175
Cleveland, OH	Dallas/Ft.Worth, TX	1,021	256
Baltimore, MD	Columbus, OH	336	121
Boston, MA	Covington, KY	752	252
Kansas City, MO	San Diego, CA	1,333	206
Milwaukee, WI	Phoenix, AZ	1,460	167
Portland, OR	Washington, DC	2,350	308
Phoenix, AZ	San Jose, CA	621	152
Baltimore, MD	St. Louis, MO	737	175
Houston, TX	Orlando, FL	853	191
Houston, TX	Seattle, WA	1,894	231
Burbank, CA	New York, NY	2,465	251
Atlanta, GA	San Diego, CA	1,891	291
Minneapolis, MN	New York, NY	1,028	260
Atlanta, GA	West Palm Beach, FL	545	123
Kansas City, MO	Seattle, WA	1,489	211
Baltimore, MD	Portland, ME	452	139
New Orleans, LA	Washington, DC	969	243

a. Draw a scatter diagram with *Distance* as the independent variable and *Fare* as the dependent variable. Is the relationship direct or indirect?
b. Compute the correlation coefficient. At the .05 significance level, is it reasonable to conclude that the correlation coefficient is greater than zero?
c. What percentage of the variation in *Fare* is accounted for by *Distance* of a flight?
d. Determine the regression equation. How much does each additional mile add to the fare? Estimate the fare for a 1,500-mile flight.
e. A traveler is planning to fly from Atlanta to London Heathrow. The distance is 4,218 miles. She wants to use the regression equation to estimate the fare. Explain why it would not be a good idea to estimate the fare for this international flight with the regression equation.

DATA ANALYTICS

62. **FILE** The North Valley Real Estate data reports information on homes on the market.
a. Let selling price be the dependent variable and size of the home the independent variable. Determine the regression equation. Estimate the selling price for a home with an area of 2,200 square feet. Determine the 95% confidence interval for all 2,200 square foot homes and the 95% prediction interval for the selling price of a home with 2,200 square feet.

b. Let days-on-the-market be the dependent variable and price be the independent variable. Determine the regression equation. Estimate the days-on-the-market of a home that is priced at $300,000. Determine the 95% confidence interval of days-on-the-market for homes with a mean price of $300,000, and the 95% prediction interval of days-on-the-market for a home priced at $300,000.

c. Can you conclude that the independent variables "days on the market" and "selling price" are positively correlated? Are the size of the home and the selling price positively correlated? Use the .05 significance level. Report the *p*-value of the test. Summarize your results in a brief report.

63. **FILE** Refer to the Baseball 2016 data, which reports information on the 2016 Major League Baseball season. Let attendance be the dependent variable and total team salary be the independent variable. Determine the regression equation and answer the following questions.

a. Draw a scatter diagram. From the diagram, does there seem to be a direct relationship between the two variables?

b. What is the expected attendance for a team with a salary of $100.0 million?

c. If the owners pay an additional $30 million, how many more people could they expect to attend?

d. At the .05 significance level, can we conclude that the slope of the regression line is positive? Conduct the appropriate test of hypothesis.

e. What percentage of the variation in attendance is accounted for by salary?

f. Determine the correlation between attendance and team batting average and between attendance and team ERA. Which is stronger? Conduct an appropriate test of hypothesis for each set of variables.

64. **FILE** Refer to the Lincolnville School bus data. Develop a regression equation that expresses the relationship between age of the bus and maintenance cost. The age of the bus is the independent variable.

a. Draw a scatter diagram. What does this diagram suggest as to the relationship between the two variables? Is it direct or indirect? Does it appear to be strong or weak?

b. Develop a regression equation. How much does an additional year add to the maintenance cost. What is the estimated maintenance cost for a 10-year-old bus?

c. Conduct a test of hypothesis to determine whether the slope of the regression line is greater than zero. Use the .05 significance level. Interpret your findings from parts (a), (b), and (c) in a brief report.